

BFS Traversal

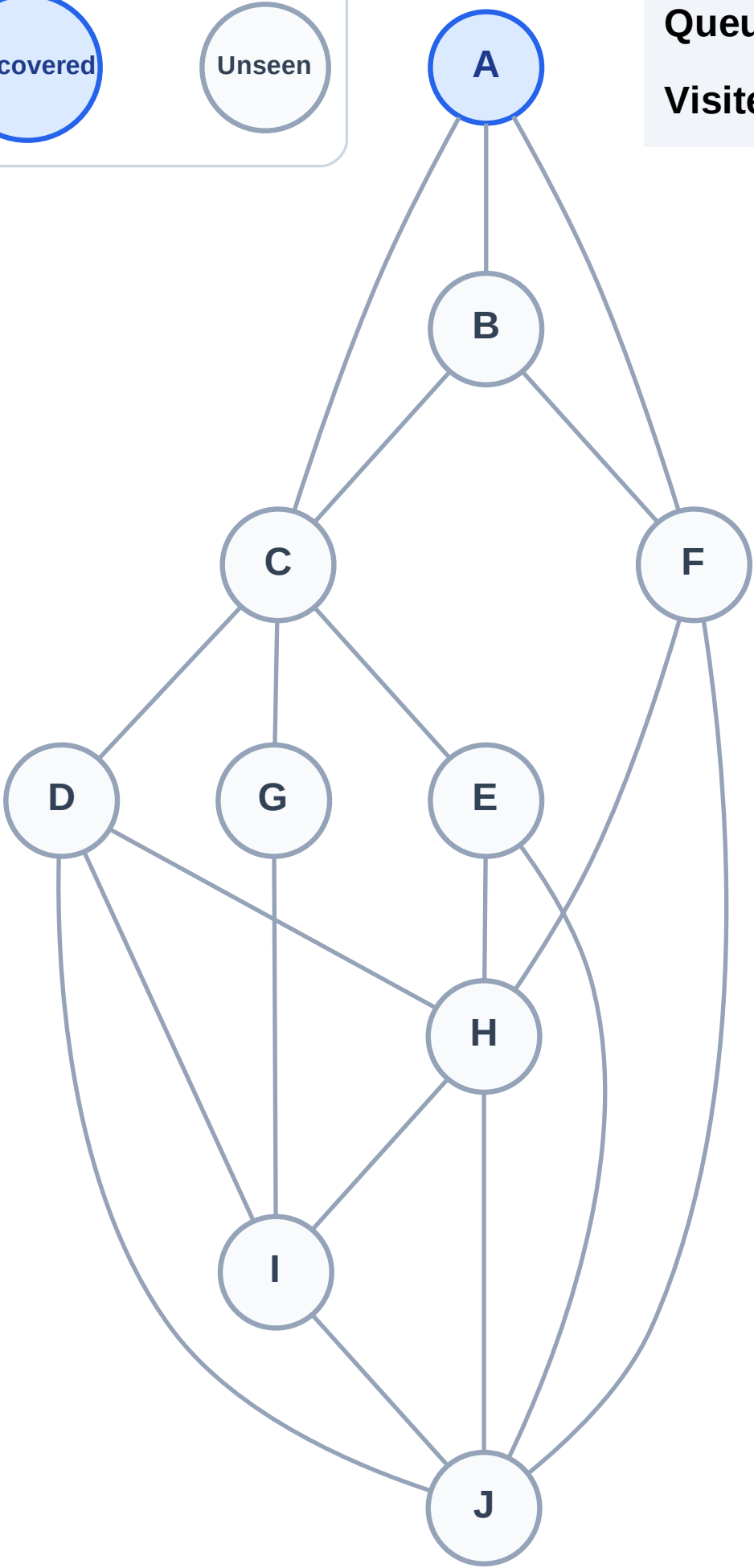
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

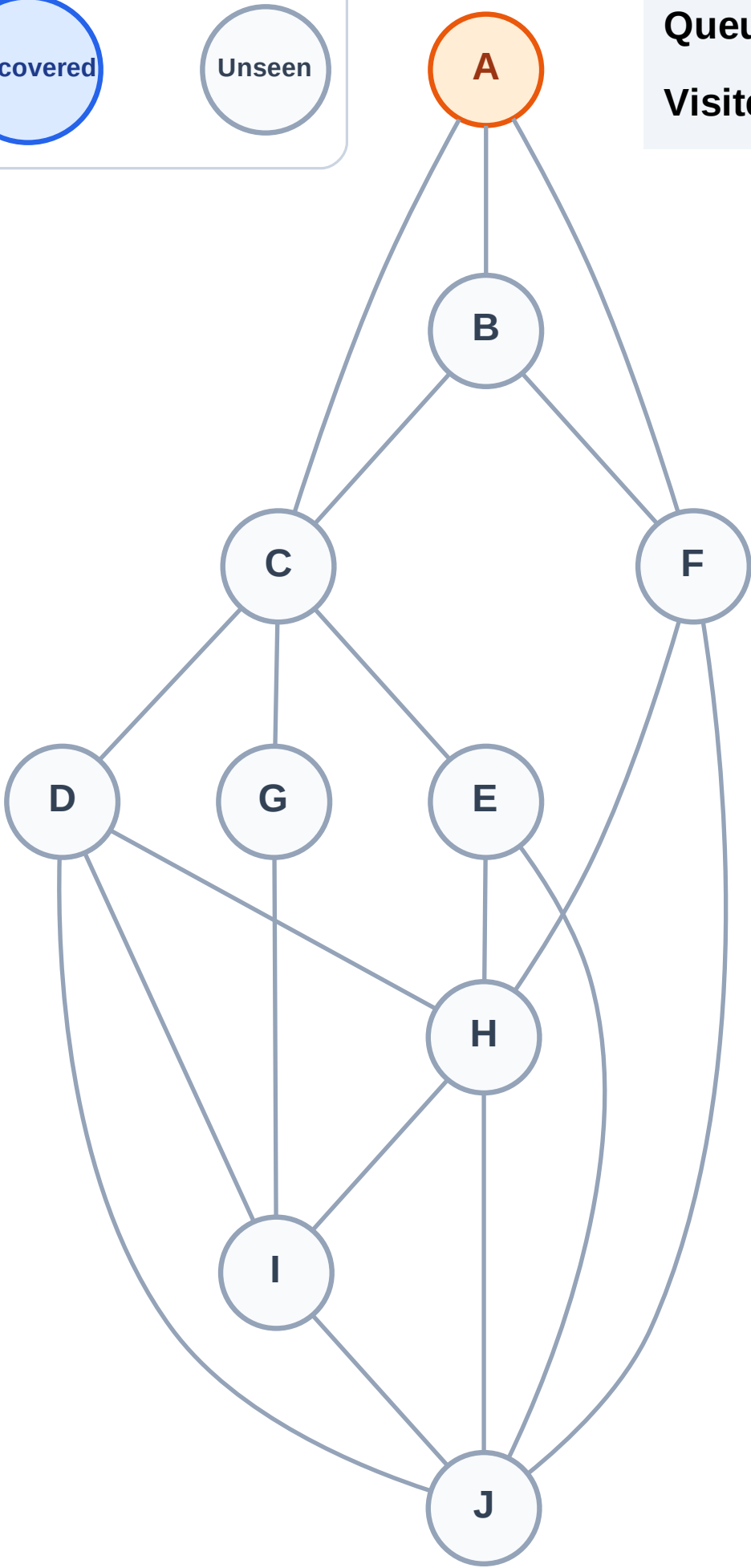
Step 2: Process node A

Legend



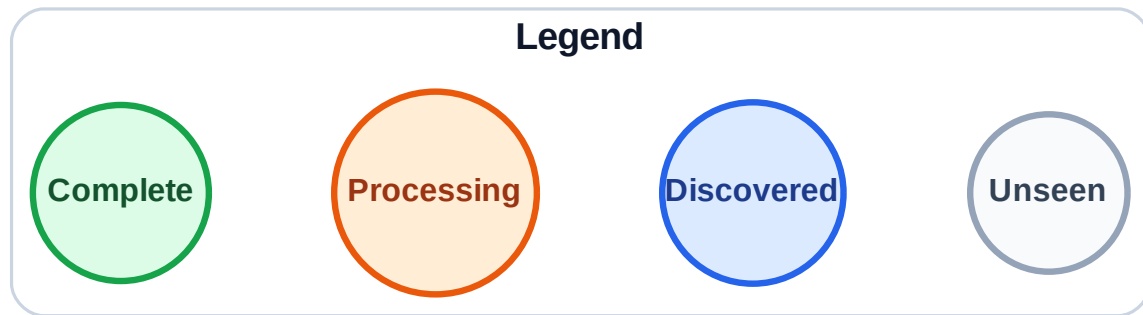
Queue: (empty)

Visited: (none)

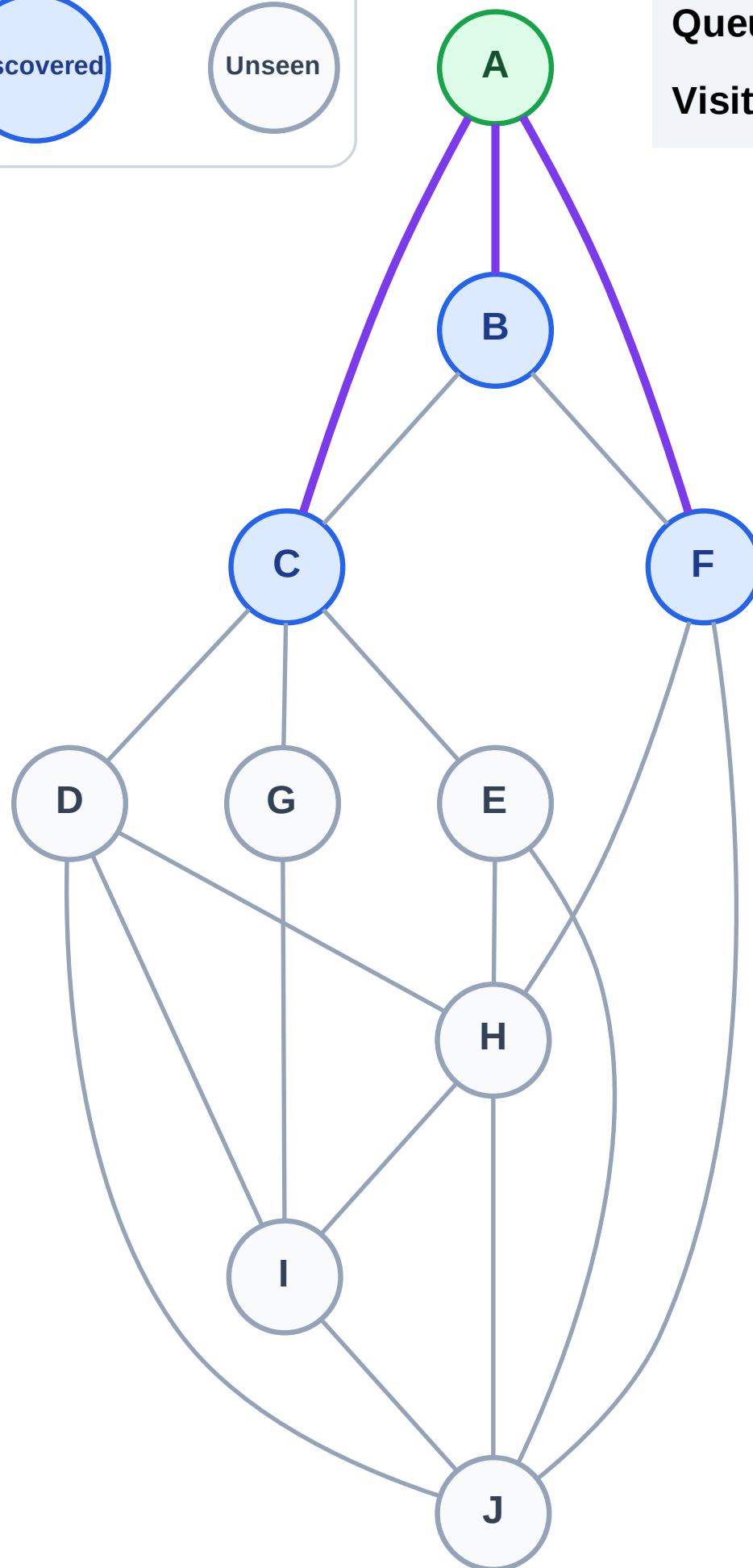


BFS Traversal

Step 3: Discover B, C, F from A



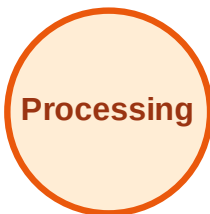
Queue: B → C → F
Visited: A



BFS Traversal

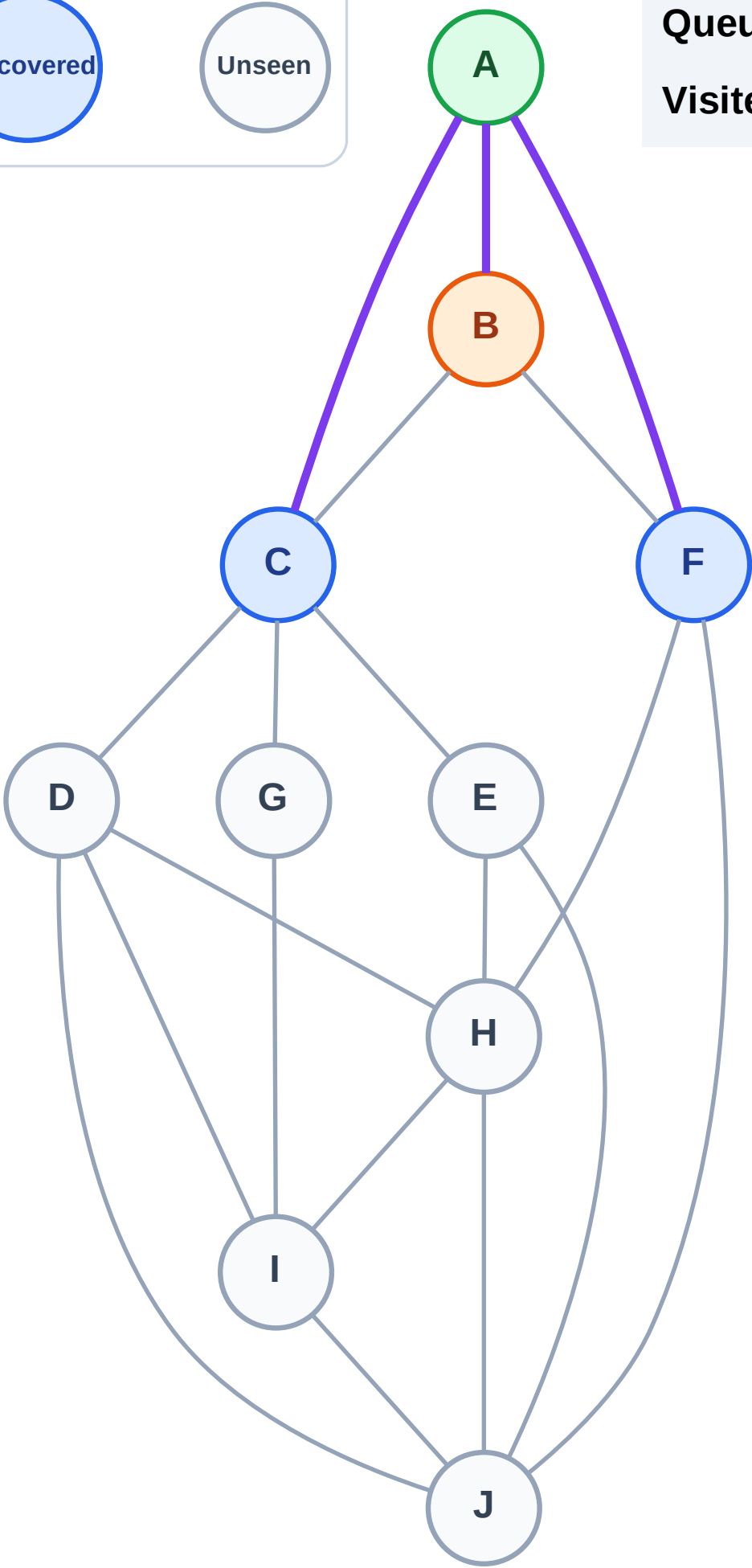
Step 4: Process node B

Legend



Queue: C → F

Visited: A



BFS Traversal

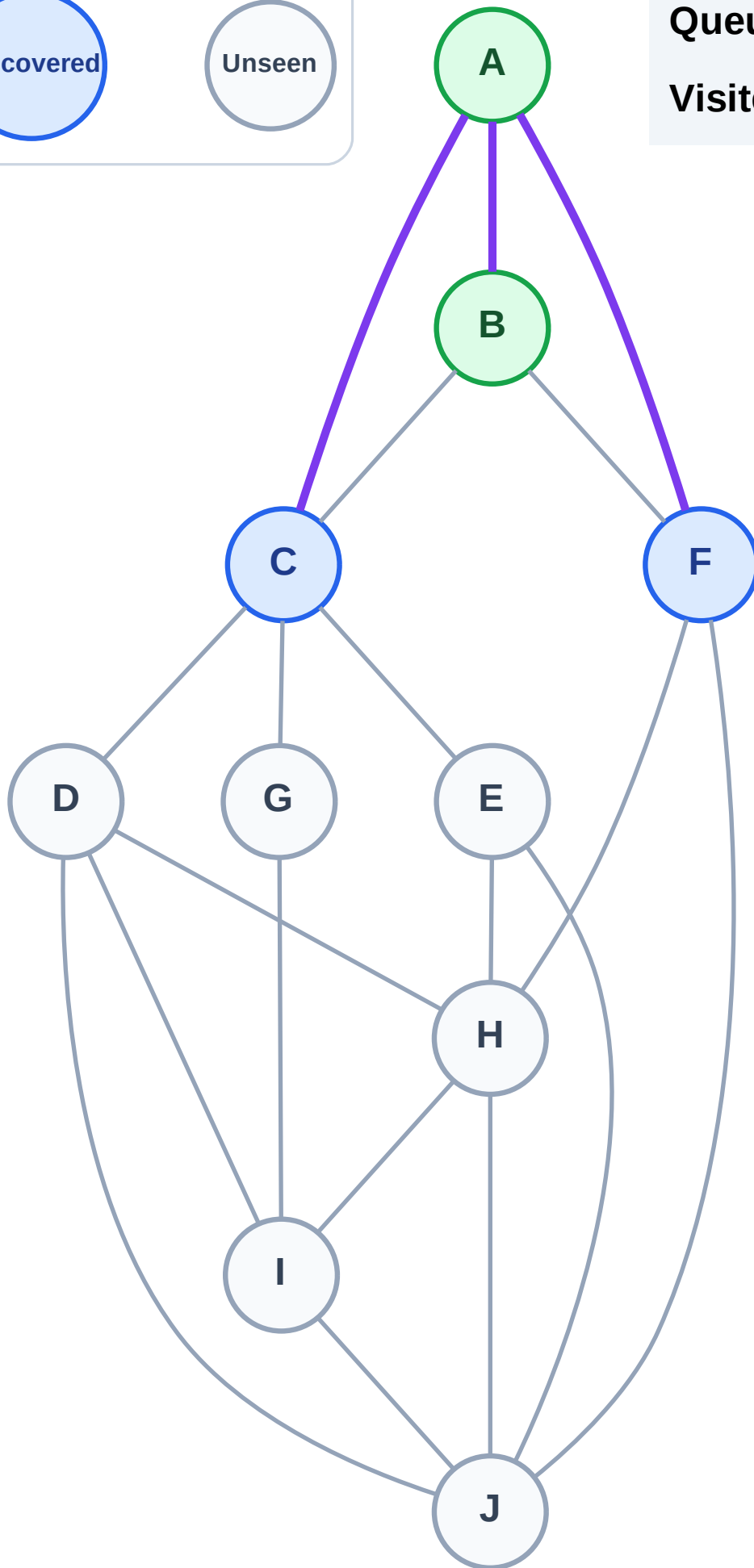
Step 5: No new neighbors from B

Legend



Queue: C → F

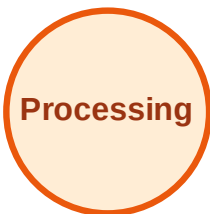
Visited: A, B



BFS Traversal

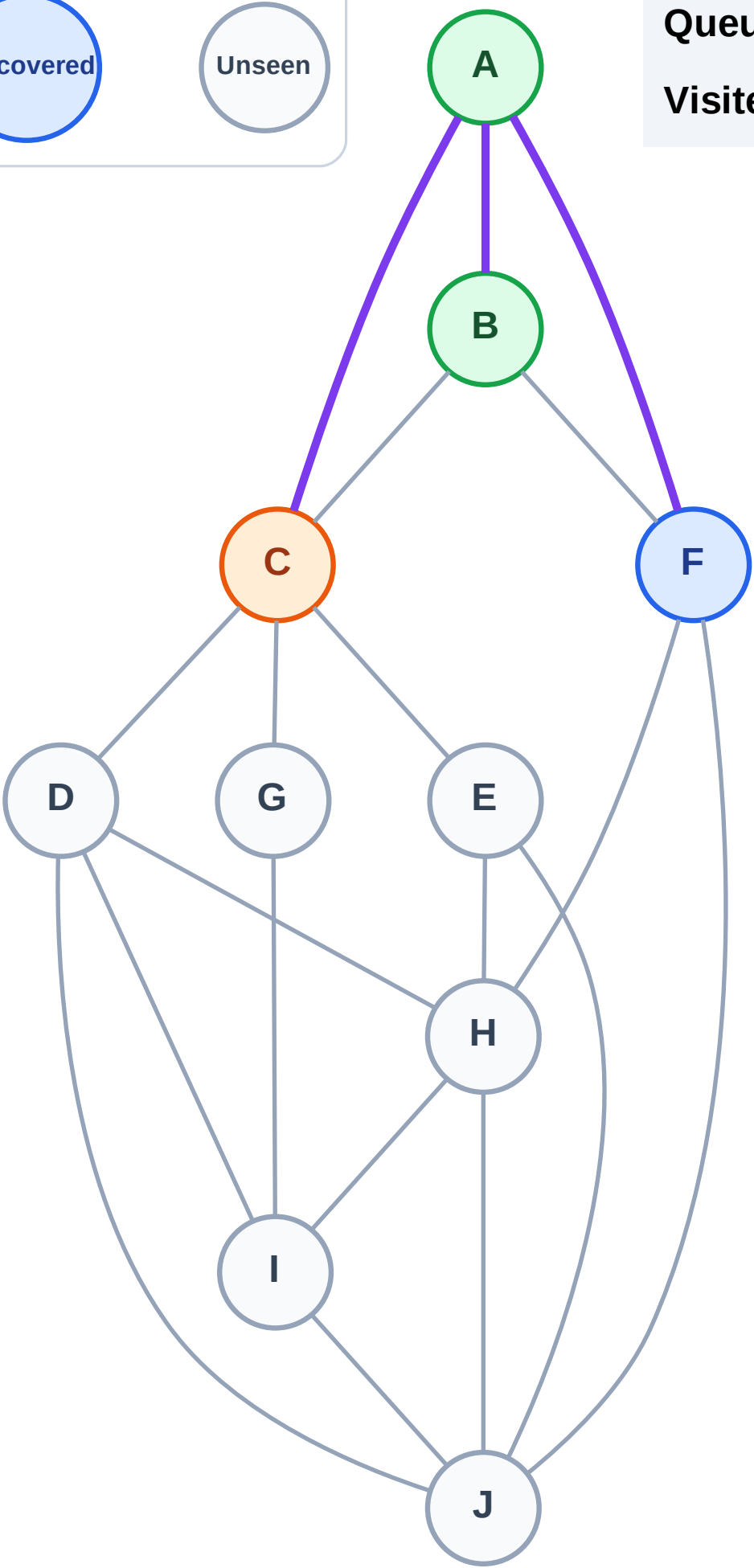
Step 6: Process node C

Legend



Queue: F

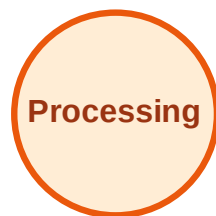
Visited: A, B



BFS Traversal

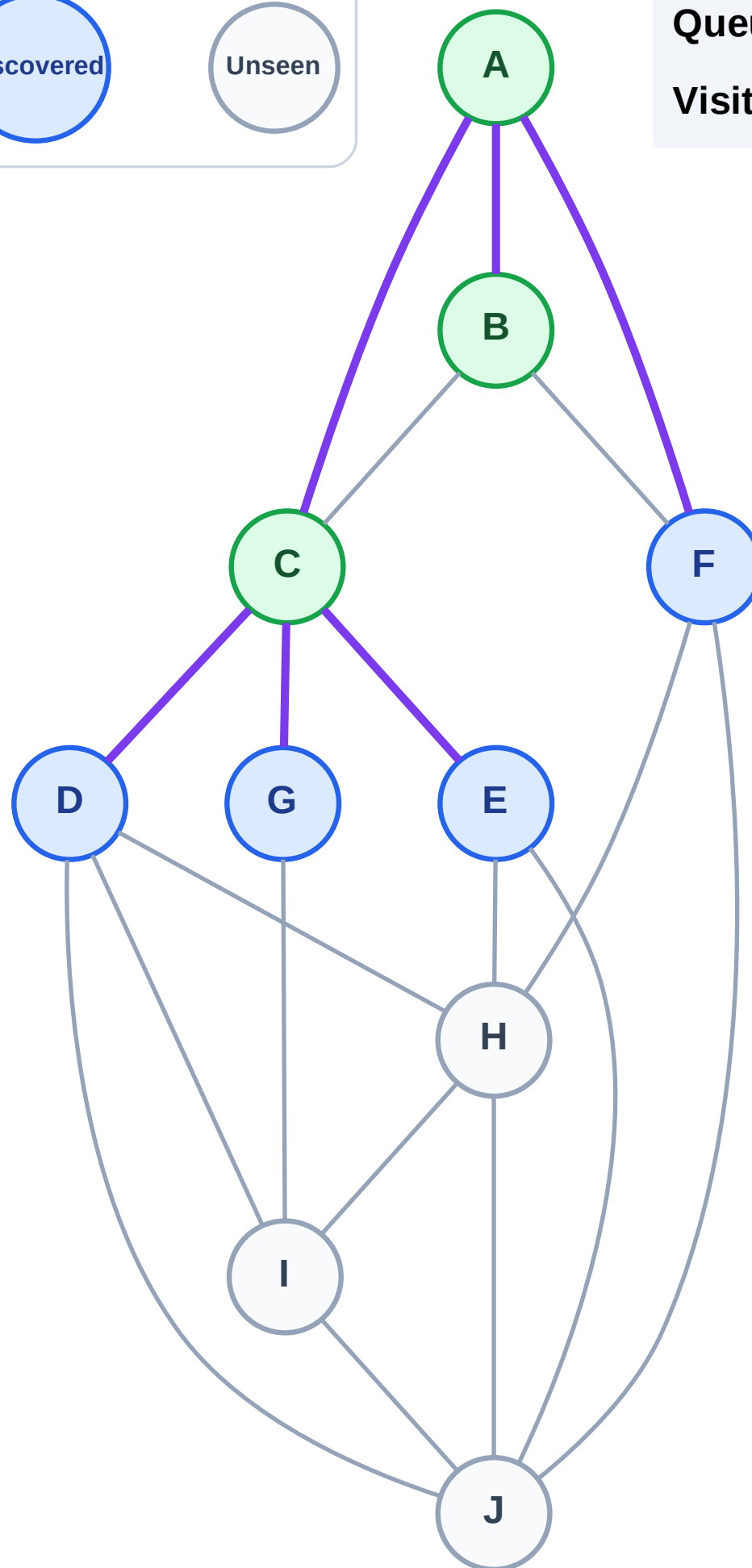
Step 7: Discover D, E, G from C

Legend



Queue: F → D → E → G

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

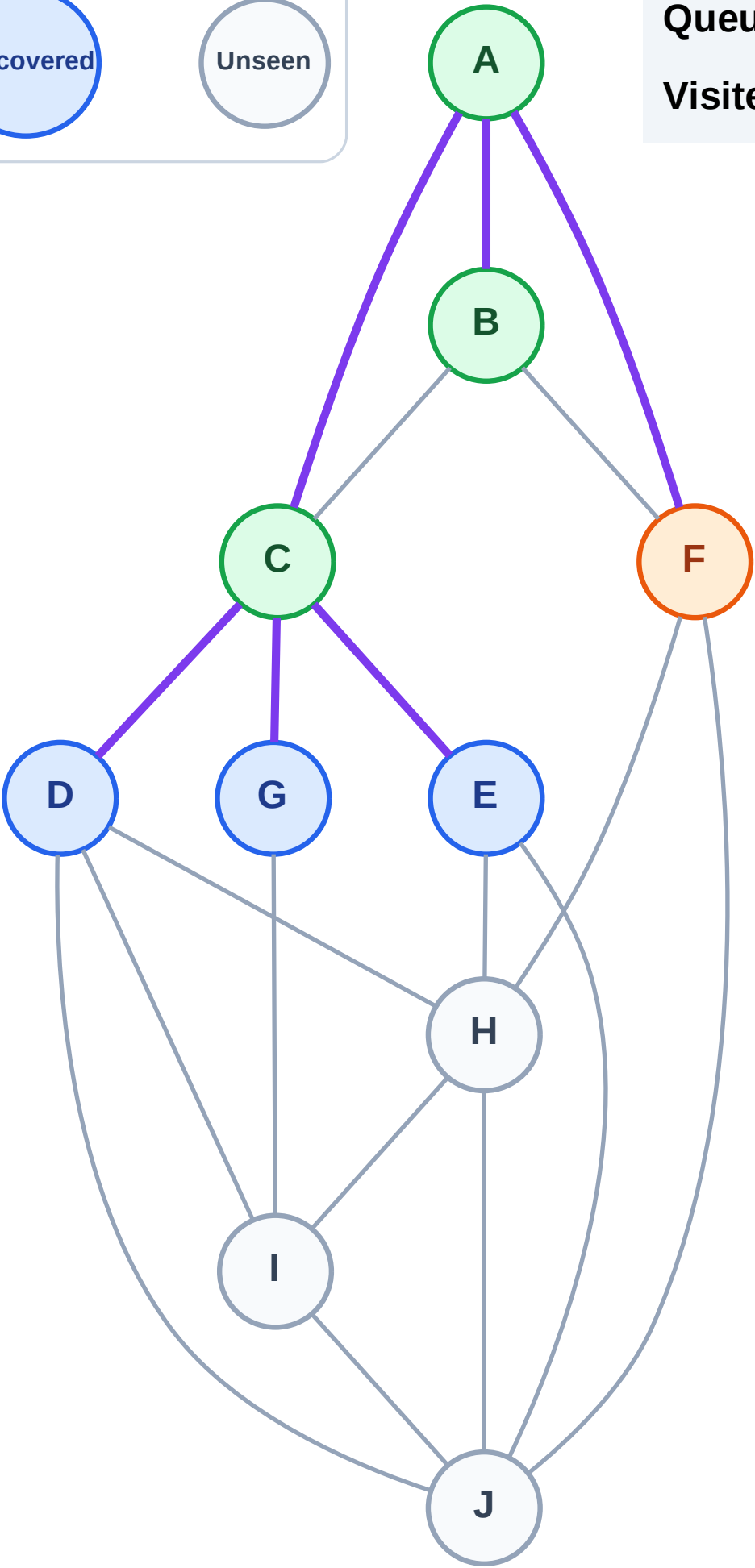
Processing

Discovered

Unseen

Queue: D → E → G

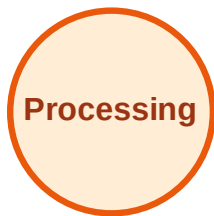
Visited: A, B, C



BFS Traversal

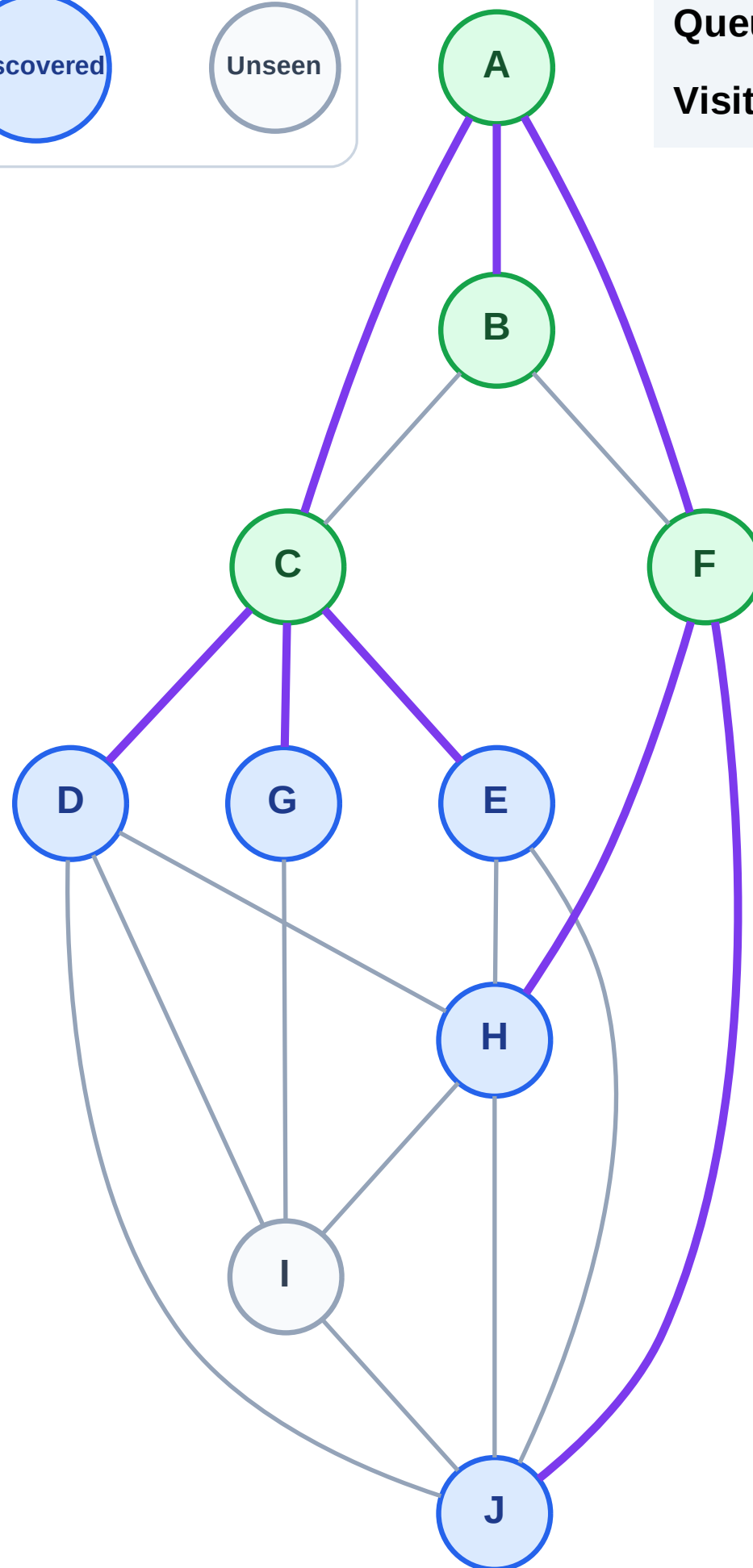
Step 9: Discover H, J from F

Legend



Queue: D → E → G → H → J

Visited: A, B, C, F



BFS Traversal

Step 10: Process node D

Legend

Complete

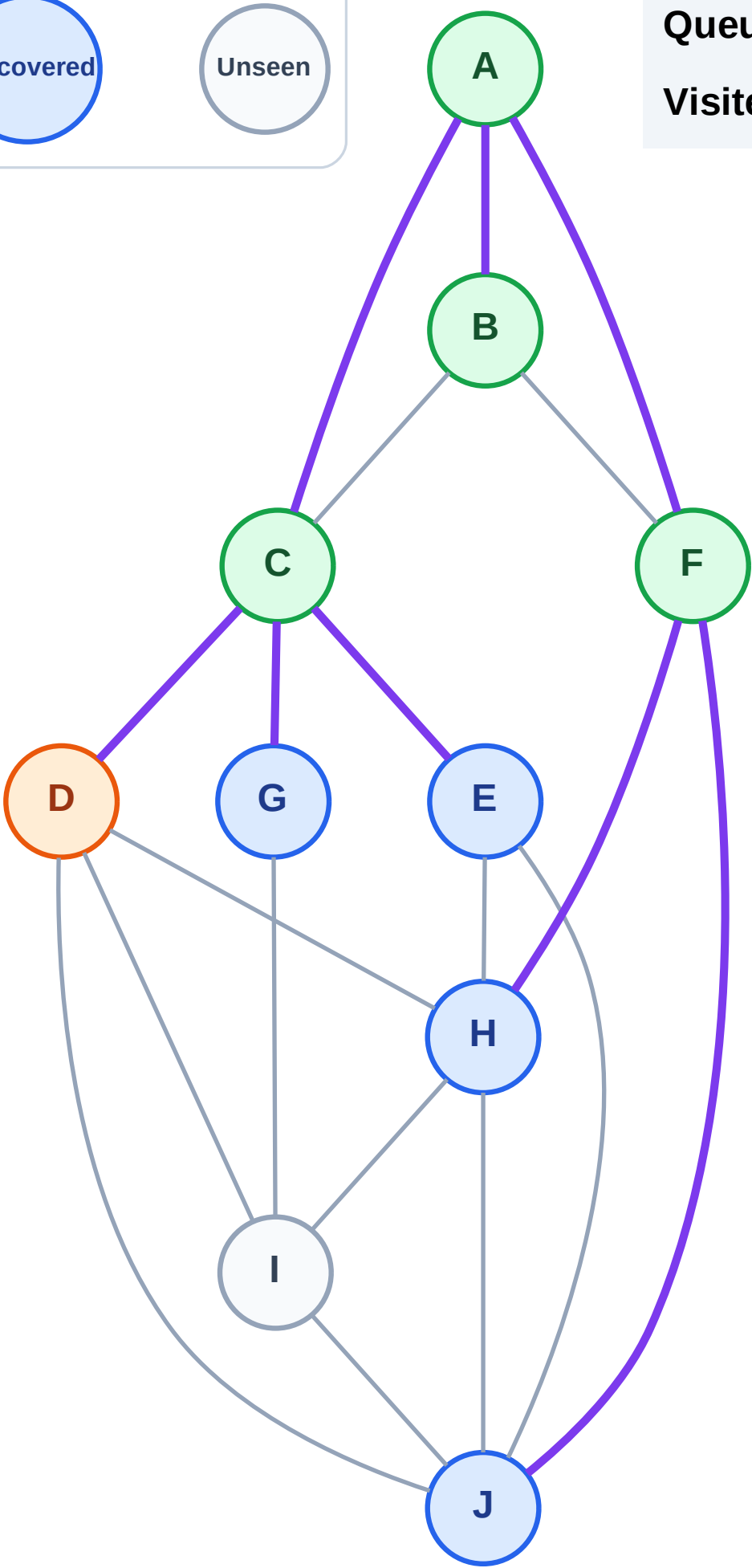
Processing

Discovered

Unseen

Queue: E → G → H → J

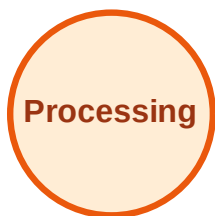
Visited: A, B, C, F



BFS Traversal

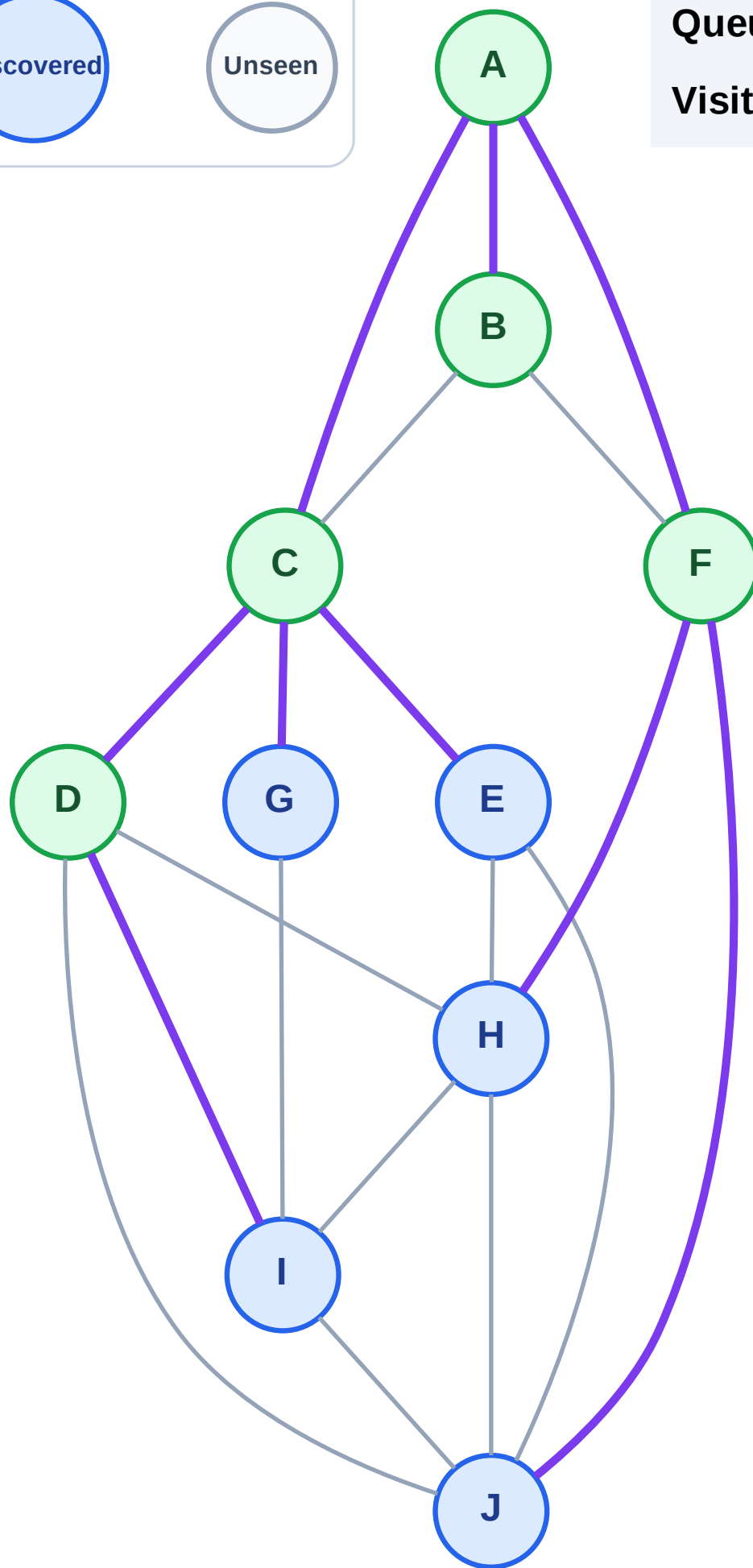
Step 11: Discover I from D

Legend



Queue: E → G → H → J → I

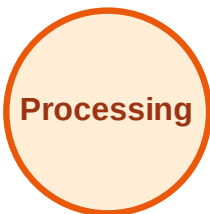
Visited: A, B, C, D, F



BFS Traversal

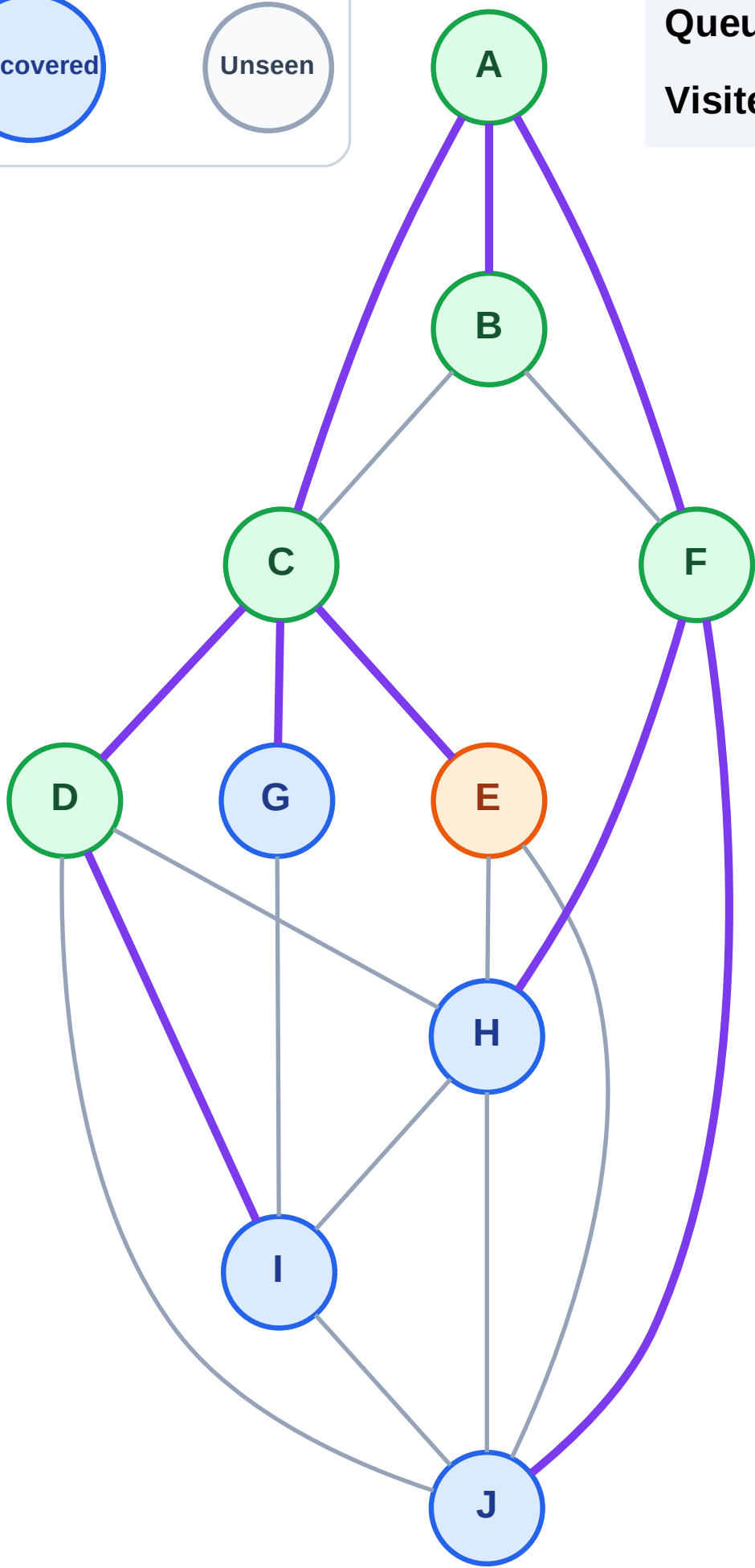
Step 12: Process node E

Legend



Queue: G → H → J → I

Visited: A, B, C, D, F



BFS Traversal

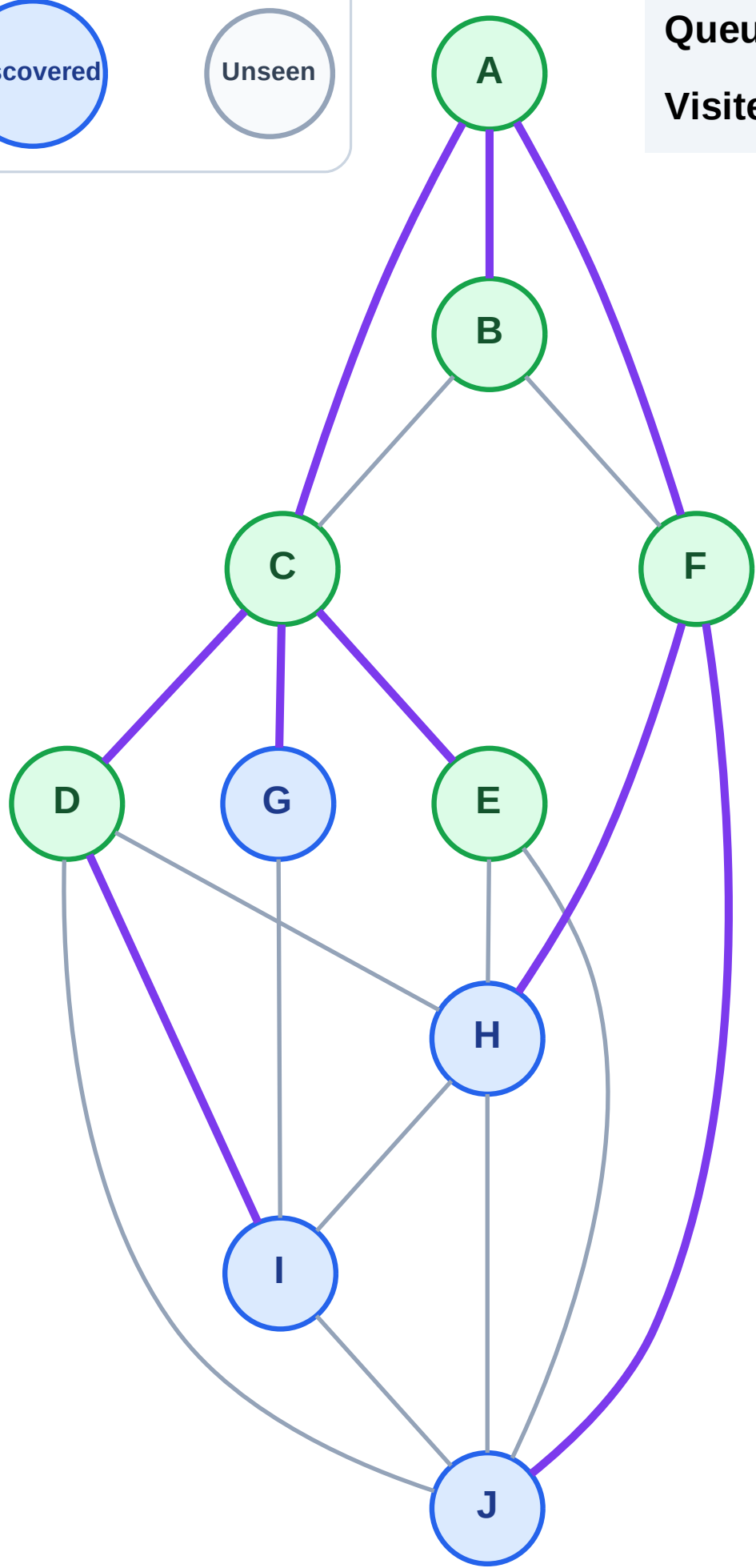
Step 13: No new neighbors from E

Legend



Queue: G → H → J → I

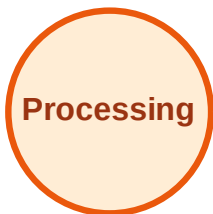
Visited: A, B, C, D, E, F



BFS Traversal

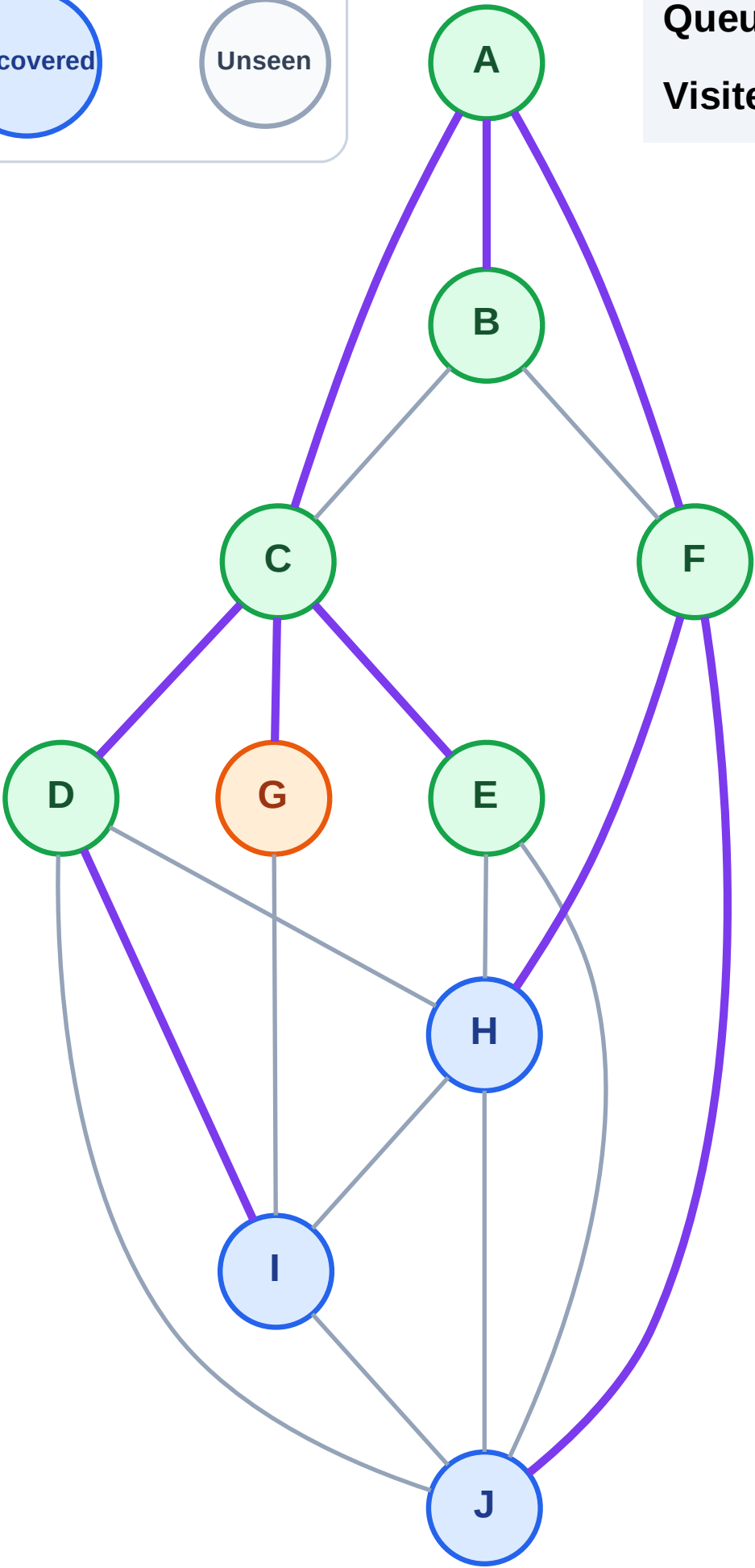
Step 14: Process node G

Legend



Queue: H → J → I

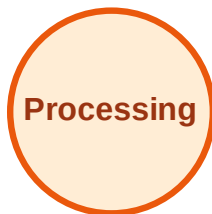
Visited: A, B, C, D, E, F



BFS Traversal

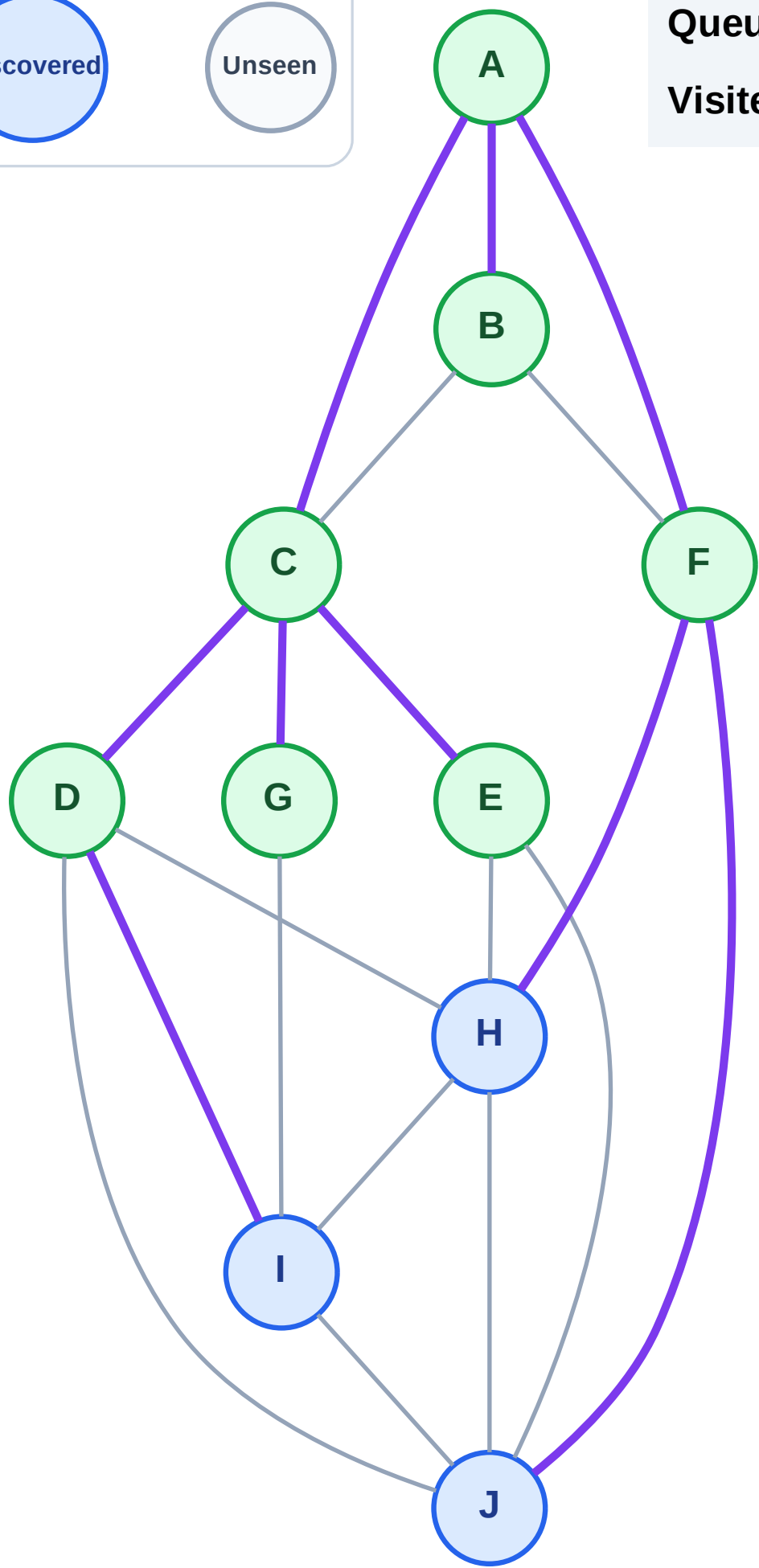
Step 15: No new neighbors from G

Legend



Queue: H → J → I

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node H

Legend

Complete

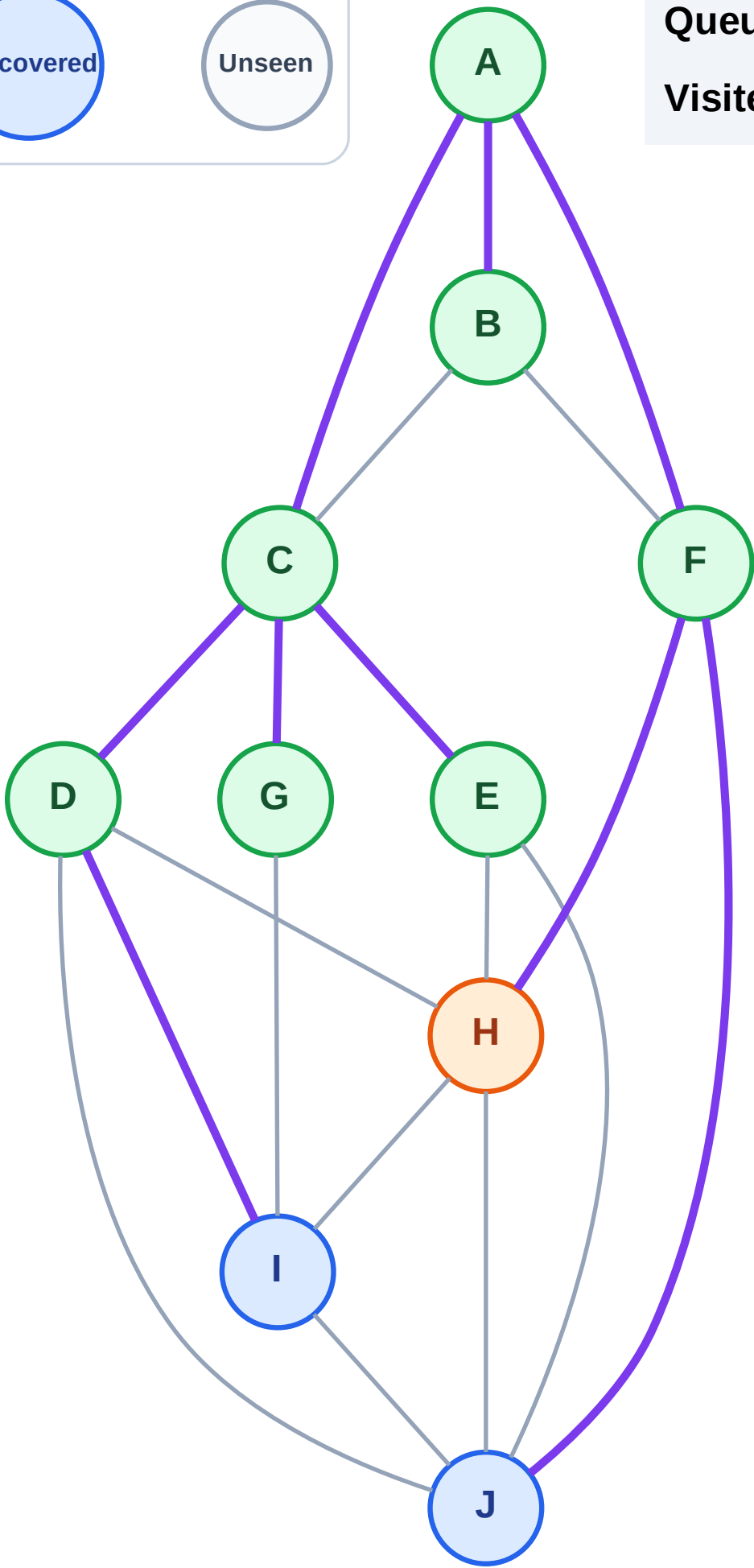
Processing

Discovered

Unseen

Queue: J → I

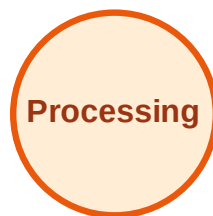
Visited: A, B, C, D, E, F, G



BFS Traversal

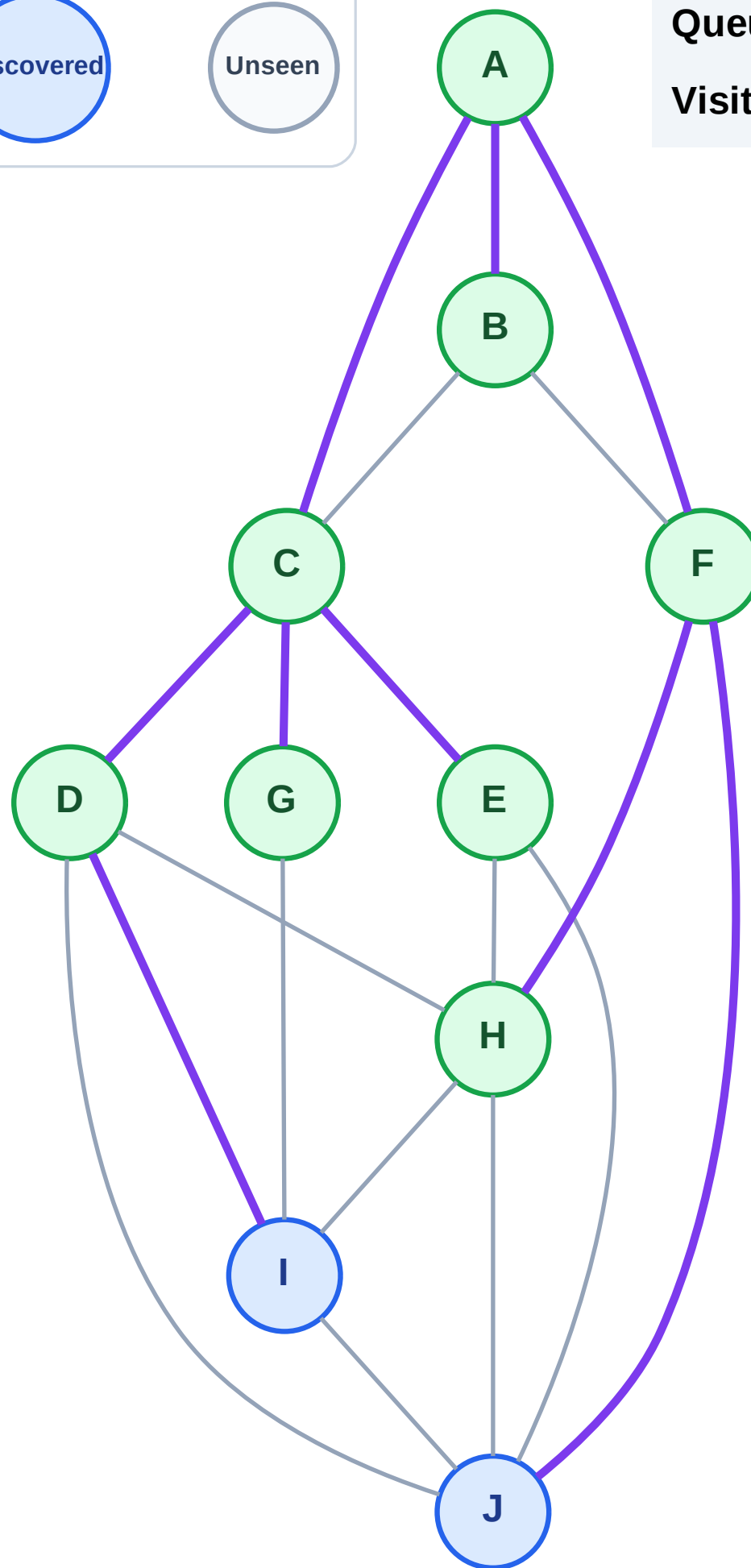
Step 17: No new neighbors from H

Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node J

Legend

Complete

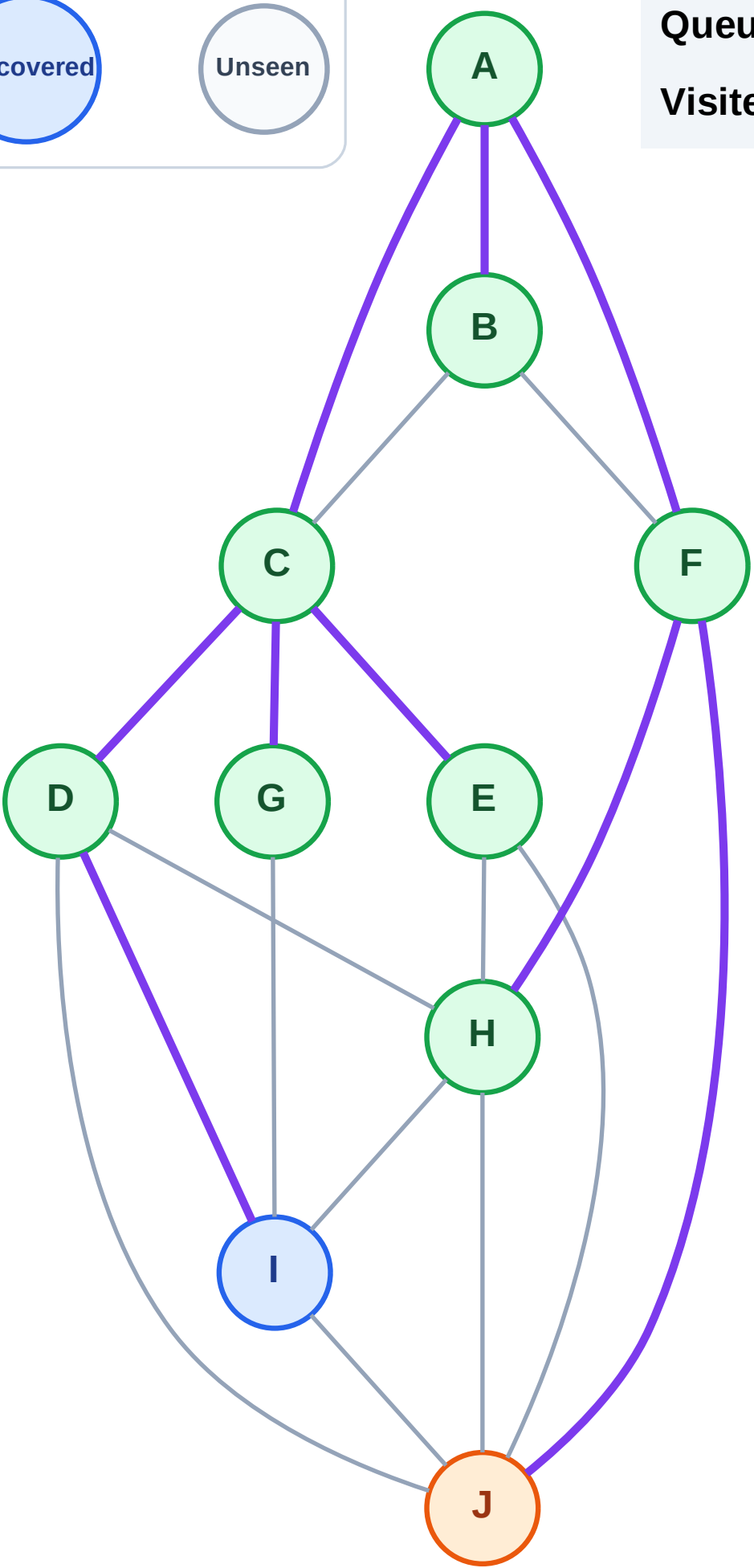
Processing

Discovered

Unseen

Queue: I

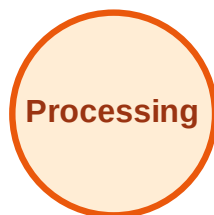
Visited: A, B, C, D, E, F, G, H



BFS Traversal

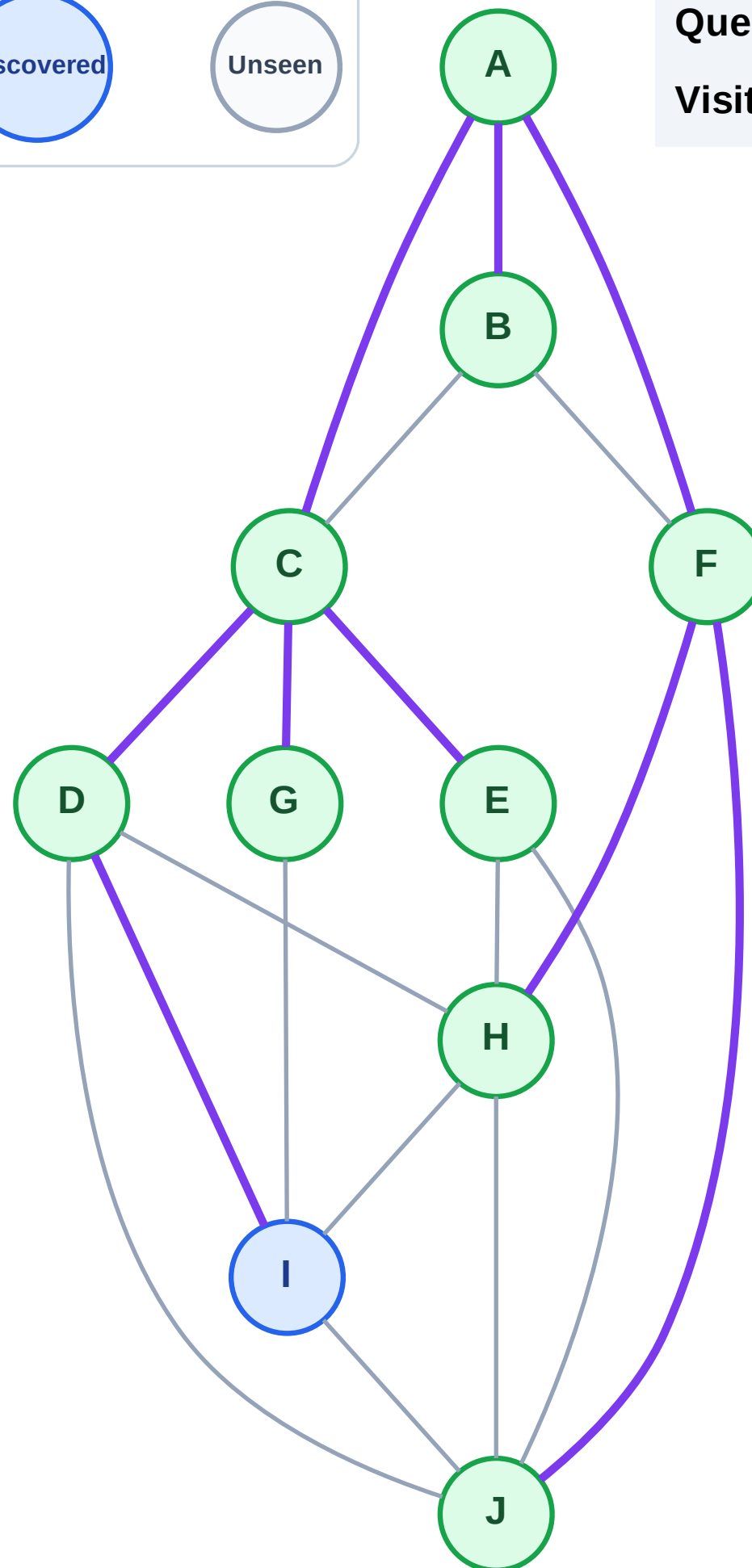
Step 19: No new neighbors from J

Legend



Queue: I

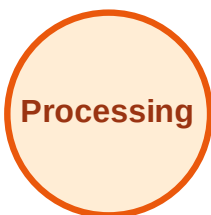
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

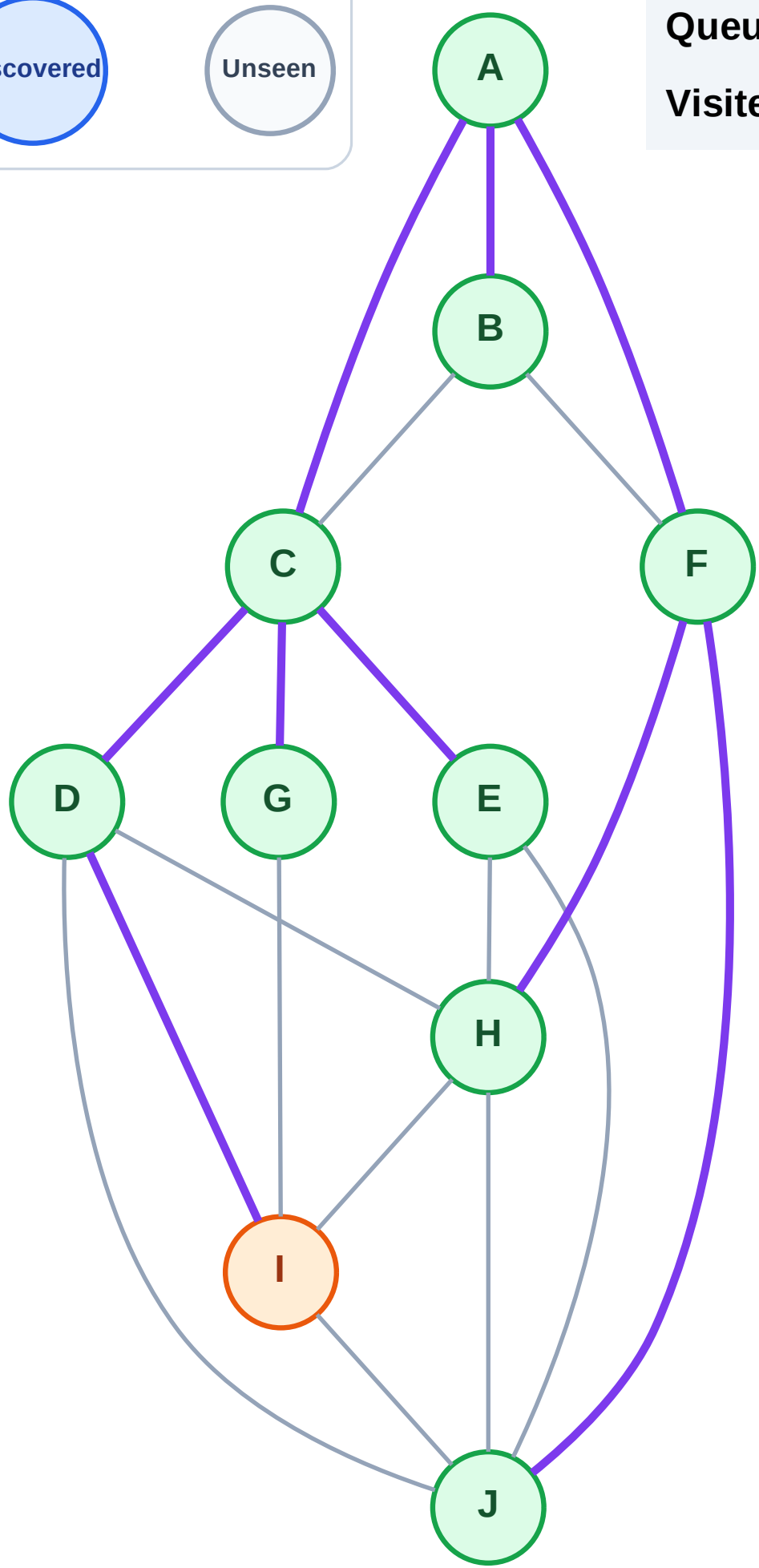
Step 20: Process node I

Legend



Queue: (empty)

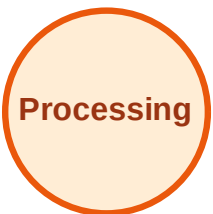
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

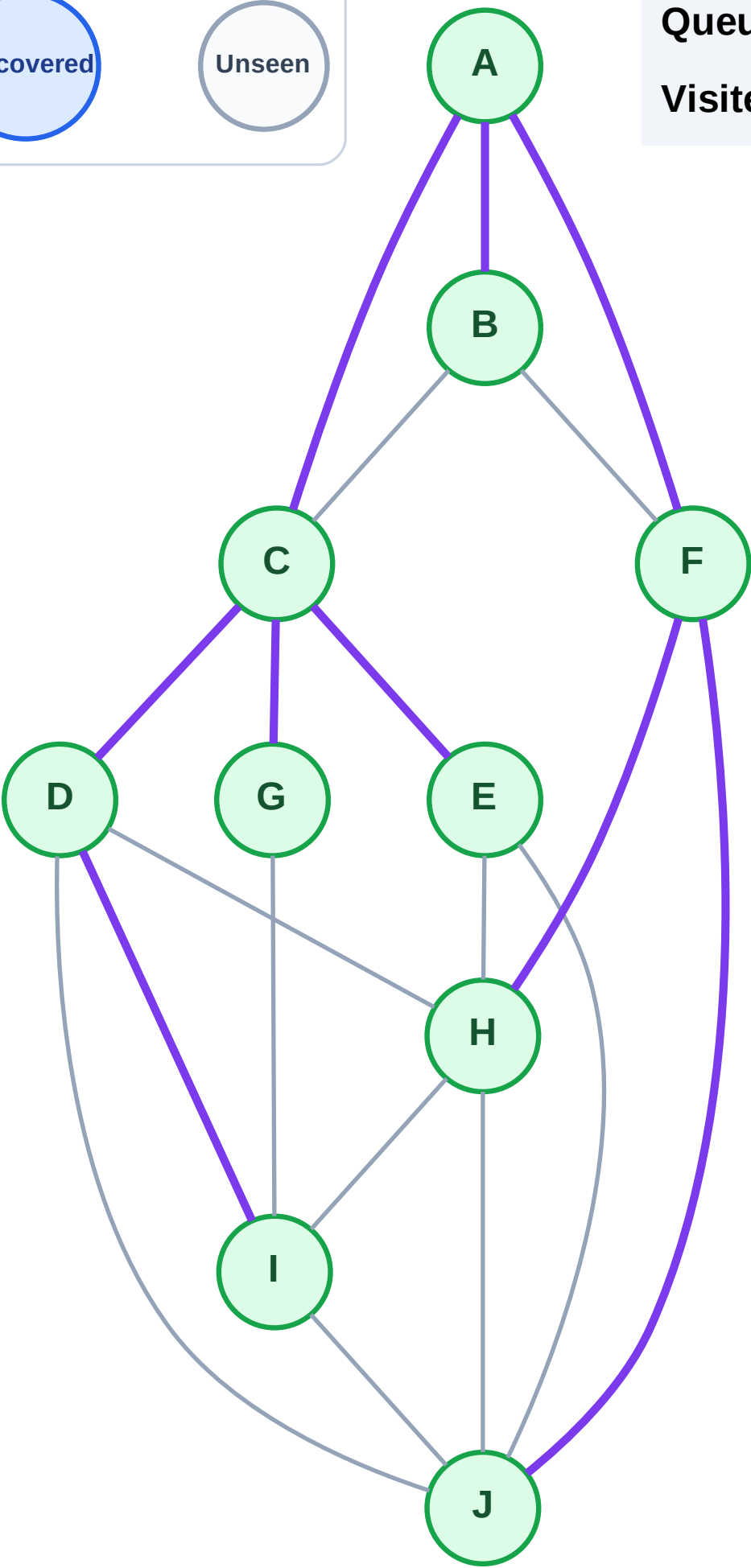
Step 21: No new neighbors from I

Legend



Queue: (empty)

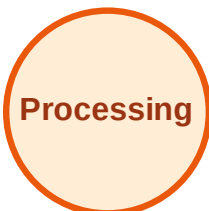
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

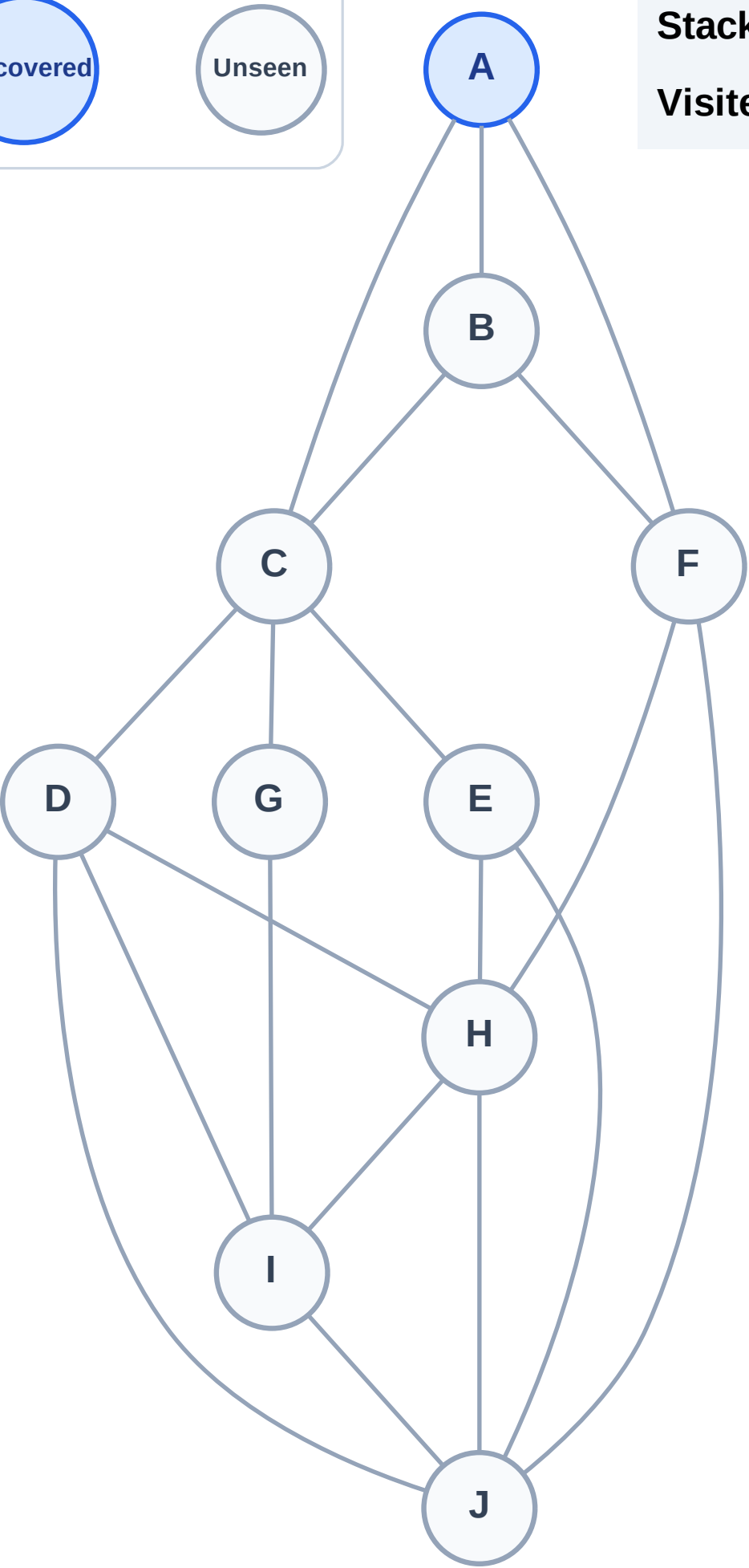
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

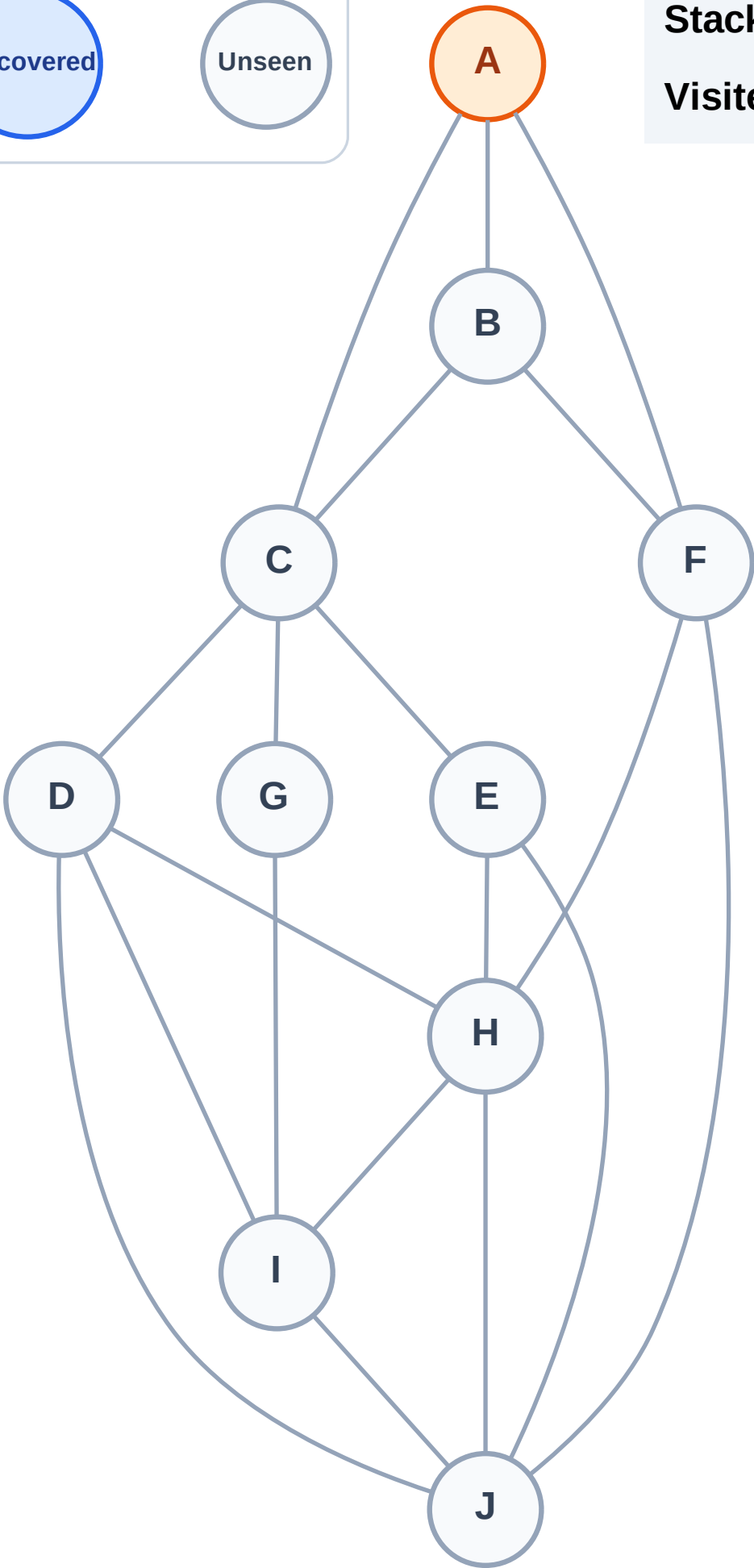
Complete

Processing

Discovered

Unseen

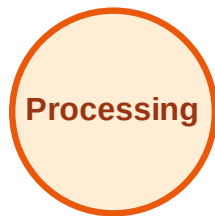
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

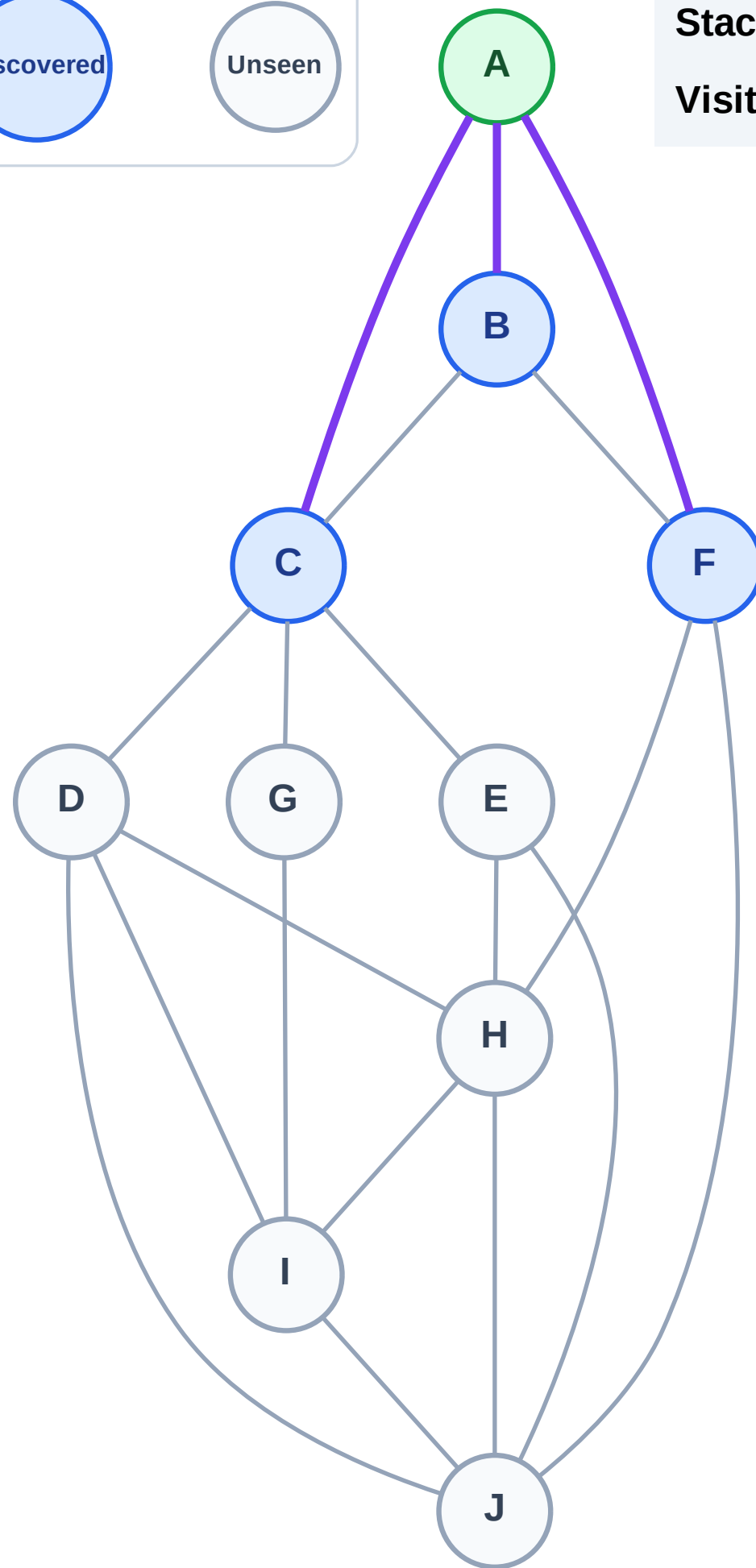
Step 3: Discover F, C, B from A

Legend



Stack (top at right): F | C | B

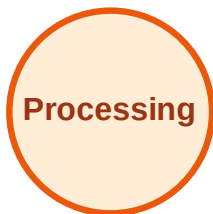
Visited: A



DFS Traversal

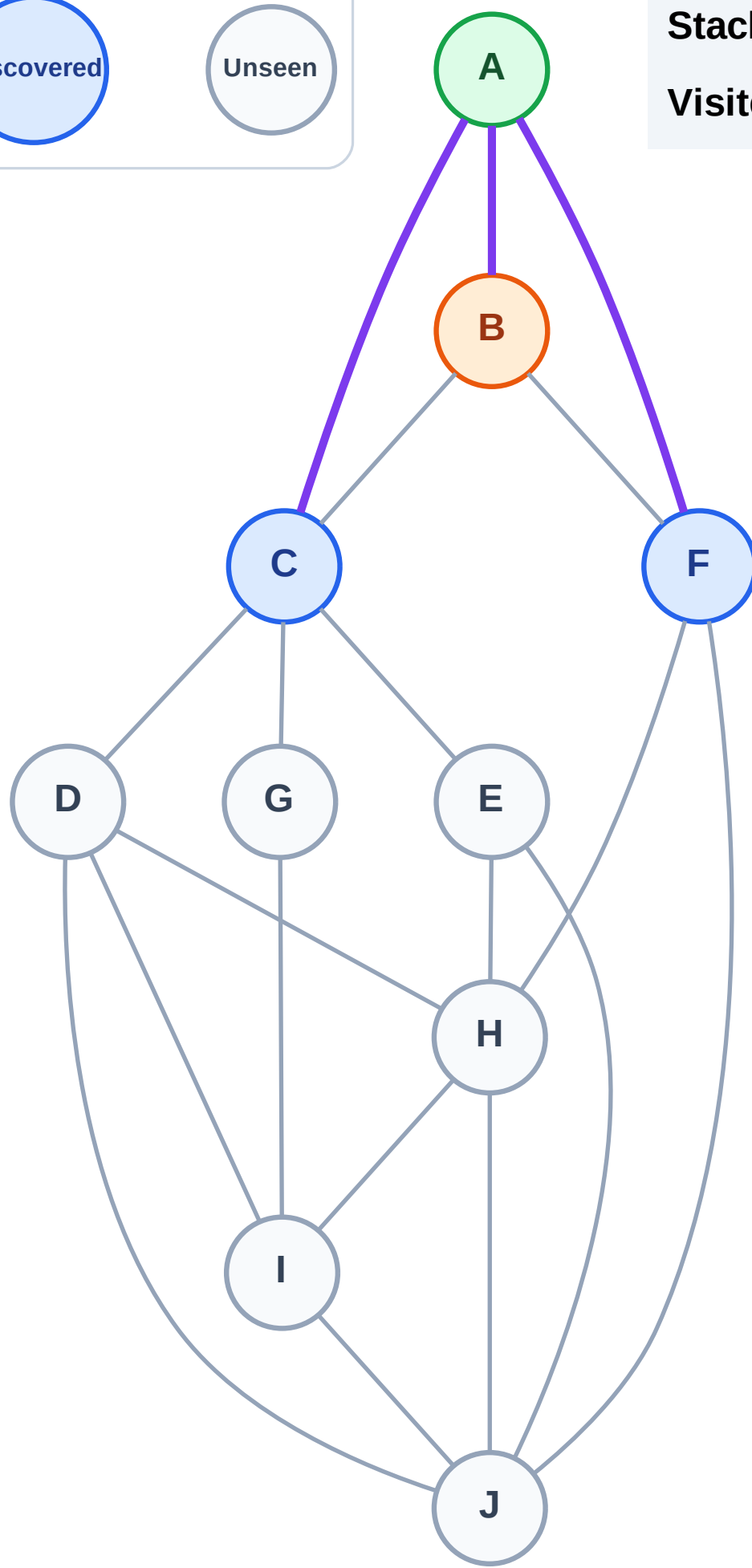
Step 4: Process node B

Legend



Stack (top at right): F | C

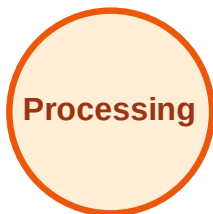
Visited: A



DFS Traversal

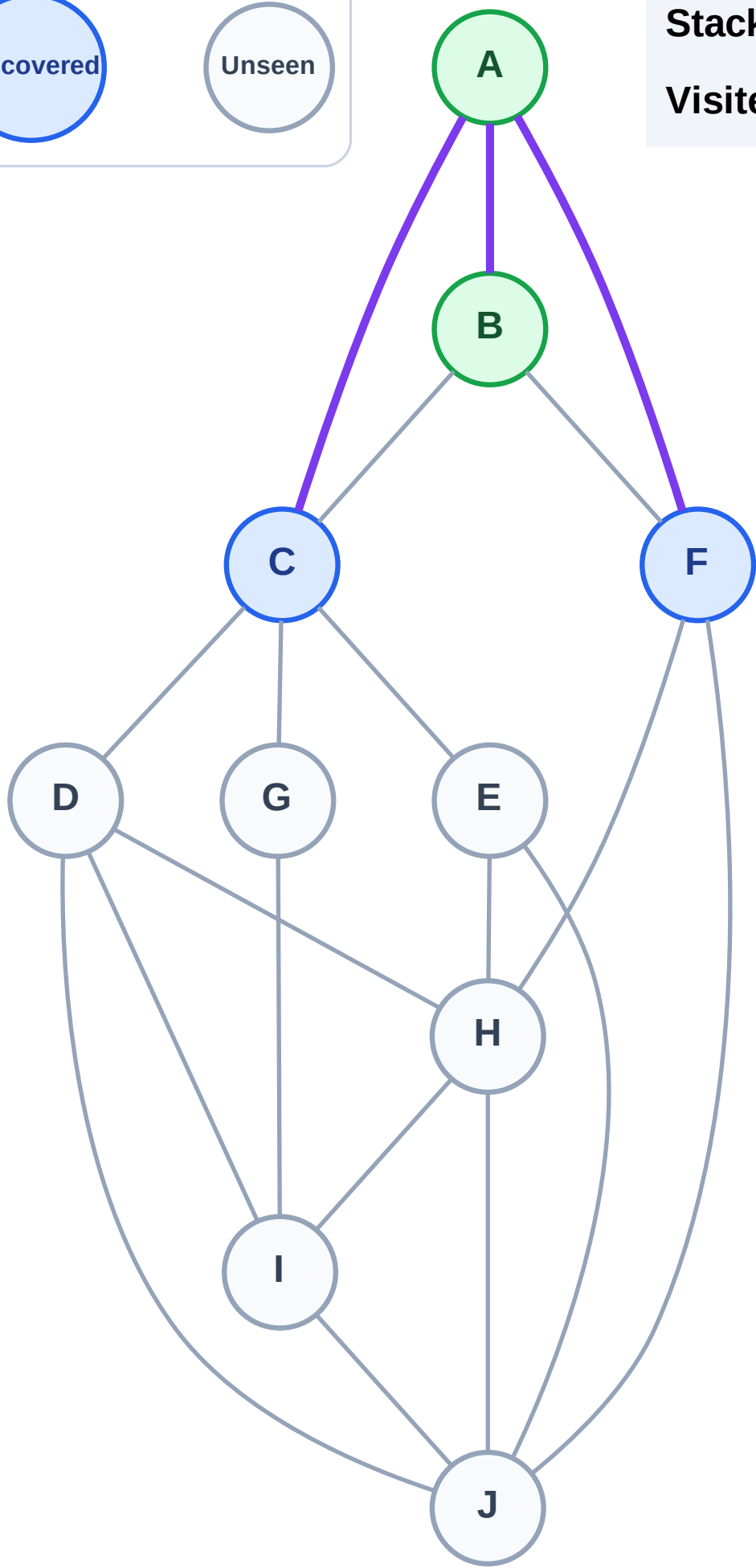
Step 5: No new neighbors from B

Legend



Stack (top at right): F | C

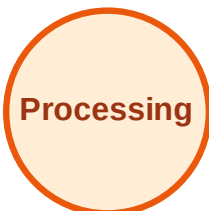
Visited: A, B



DFS Traversal

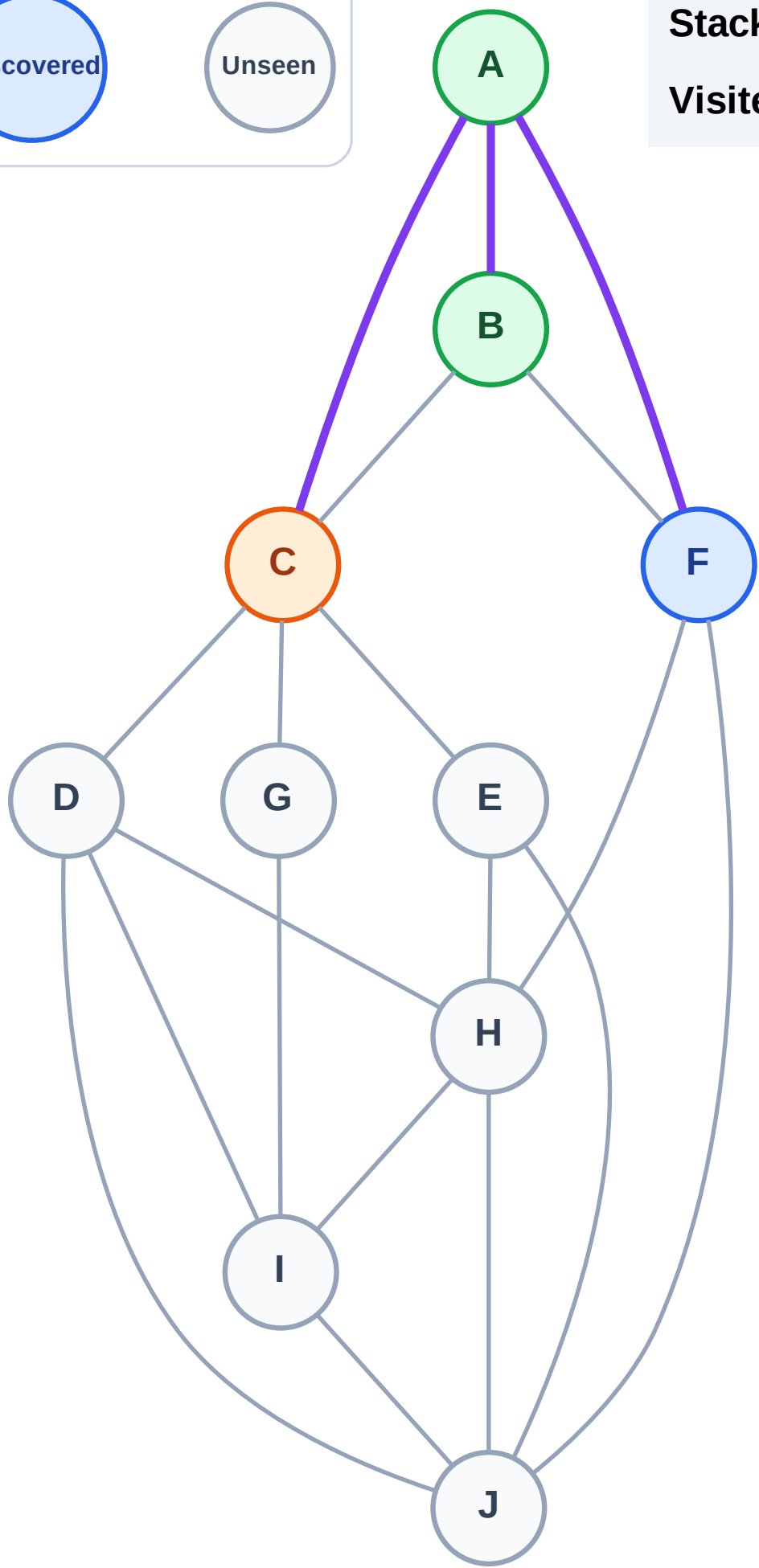
Step 6: Process node C

Legend



Stack (top at right): F

Visited: A, B



DFS Traversal

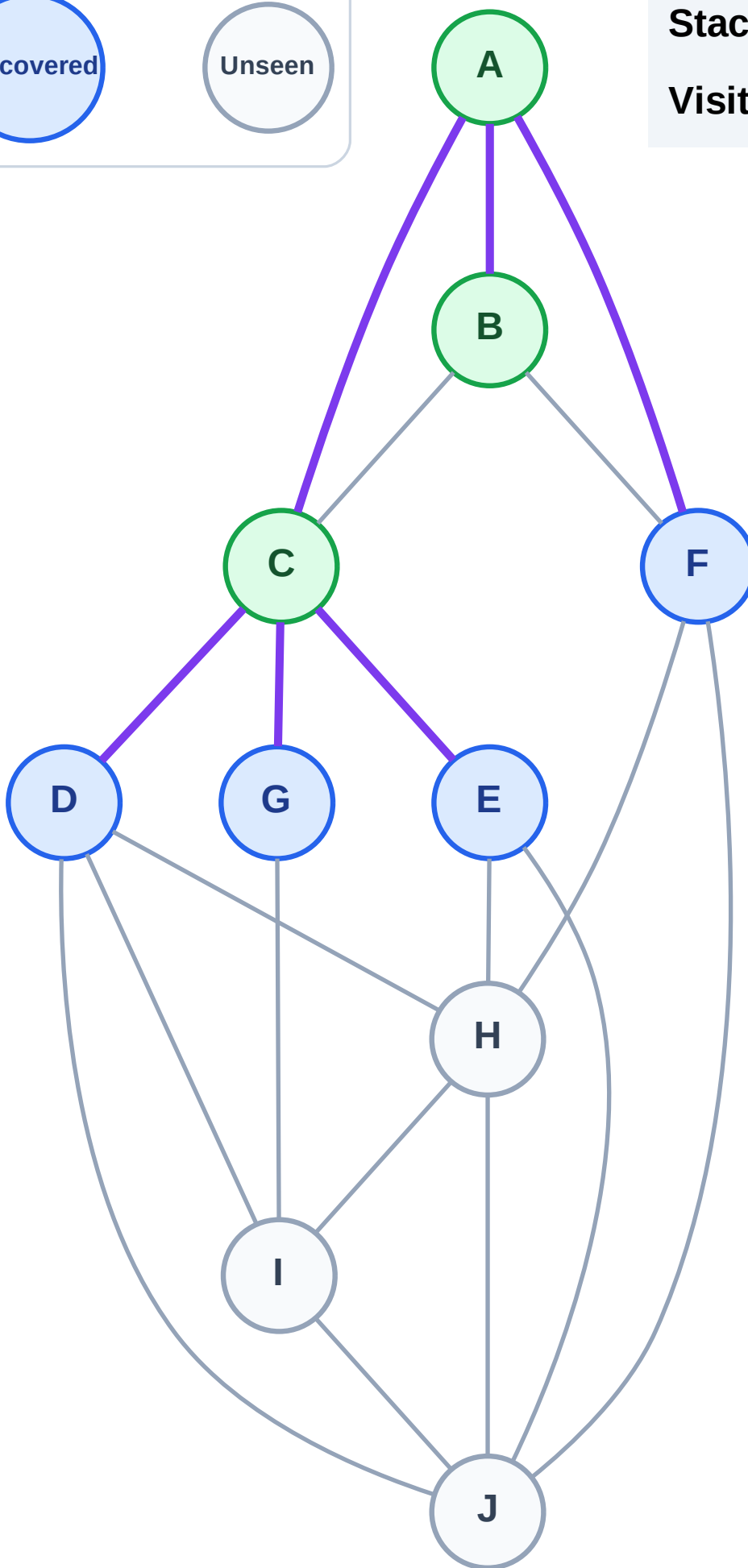
Step 7: Discover G, E, D from C

Legend



Stack (top at right): F | G | E | D

Visited: A, B, C



DFS Traversal

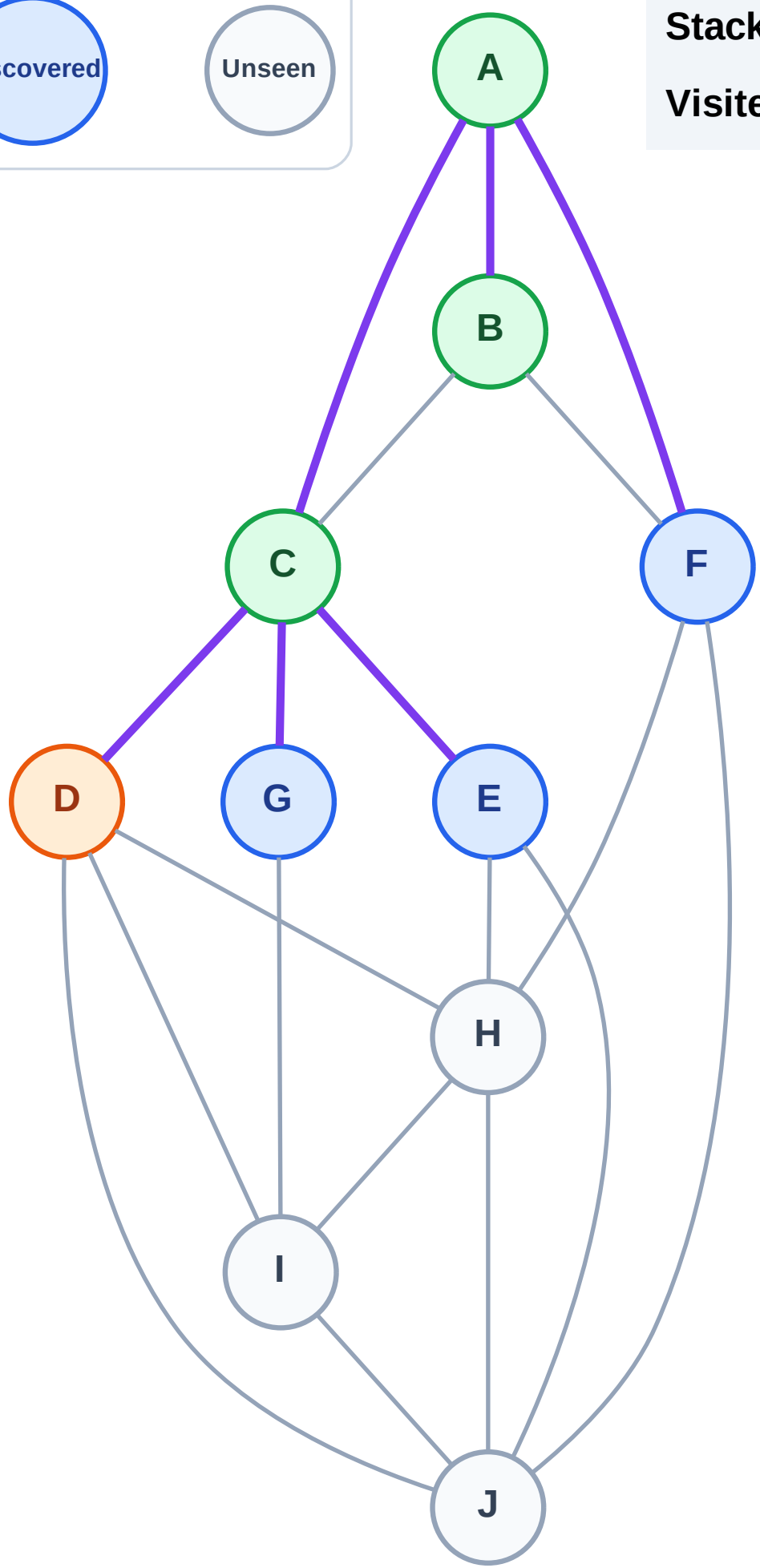
Step 8: Process node D

Legend



Stack (top at right): F | G | E

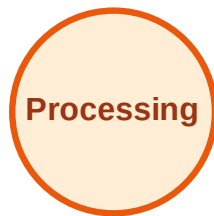
Visited: A, B, C



DFS Traversal

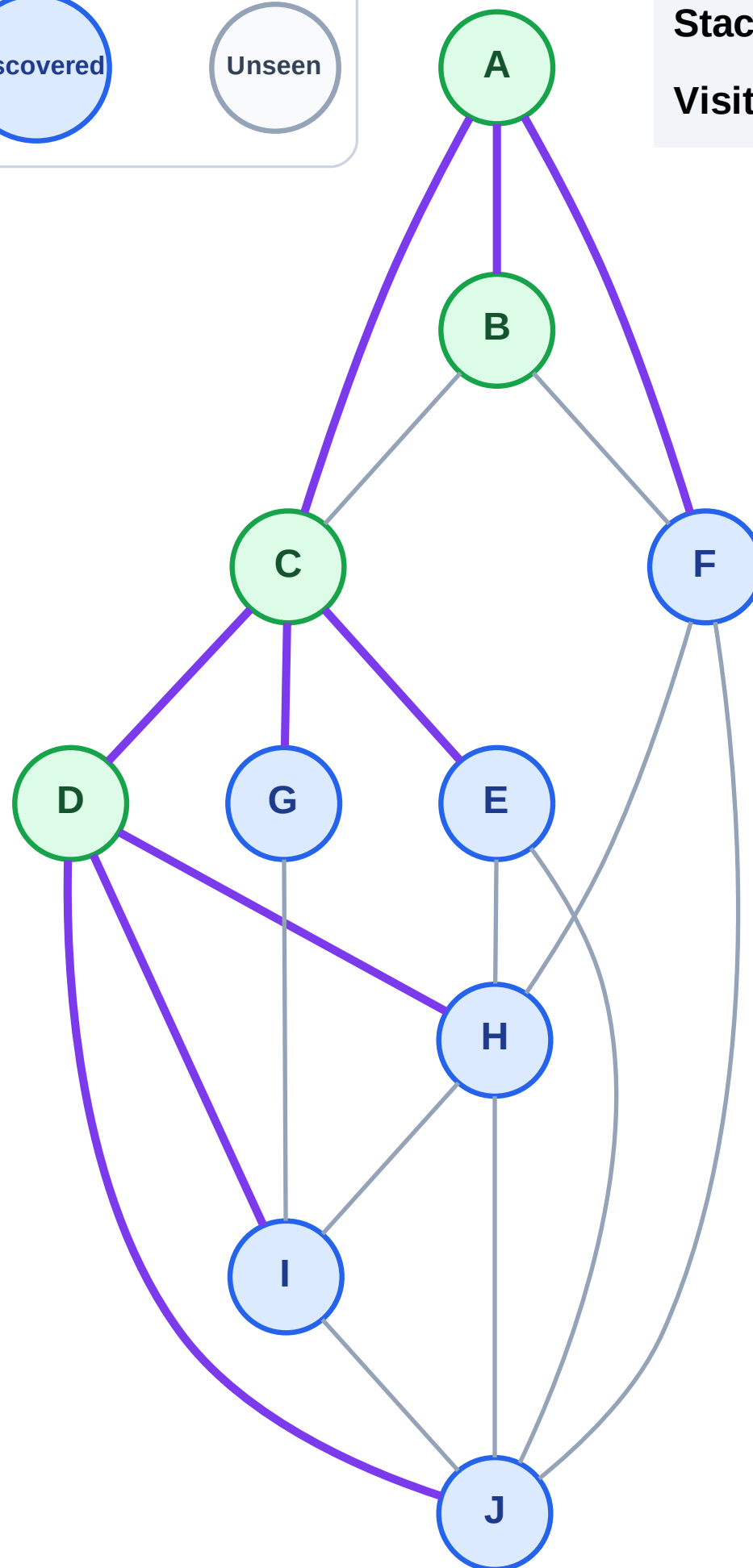
Step 9: Discover J, I, H from D

Legend



Stack (top at right): F | G | E | J | I | H

Visited: A, B, C, D



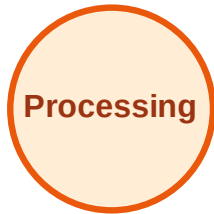
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



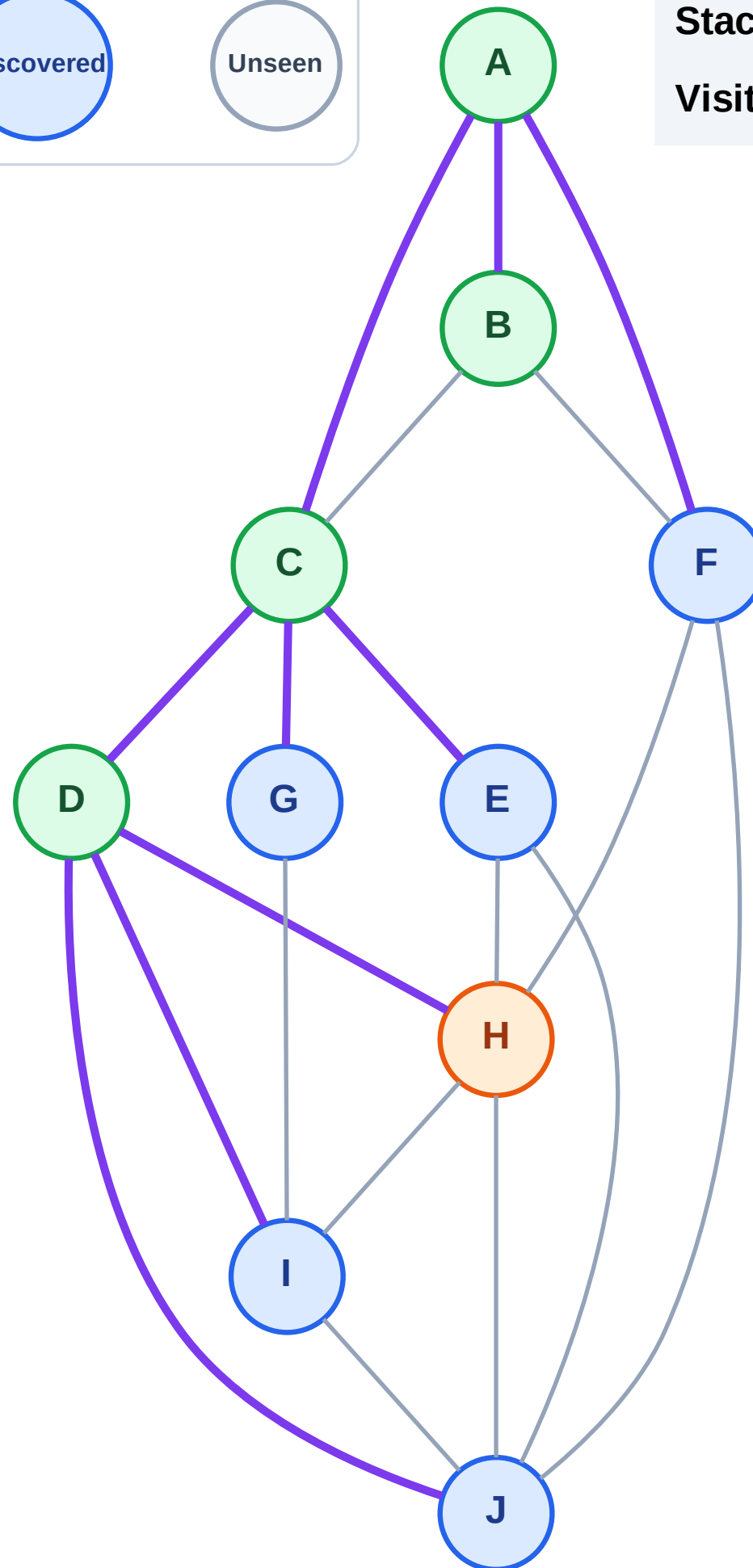
Discovered



Unseen

Stack (top at right): F | G | E | J | I

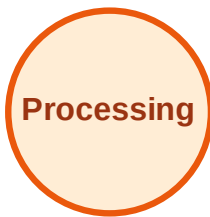
Visited: A, B, C, D



DFS Traversal

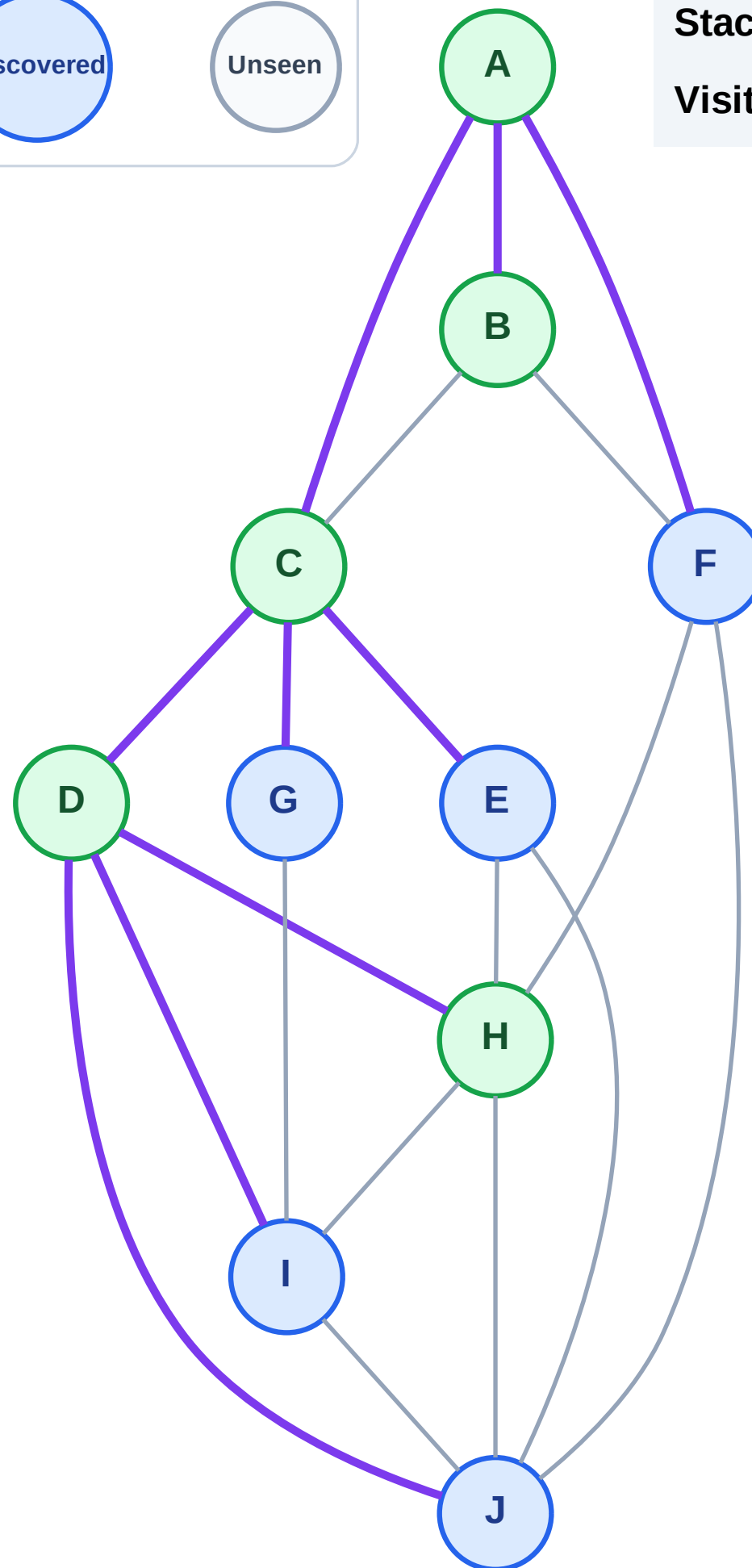
Step 11: No new neighbors from H

Legend



Stack (top at right): F | G | E | J | I

Visited: A, B, C, D, H



DFS Traversal

Step 12: Process node I

Legend

Complete

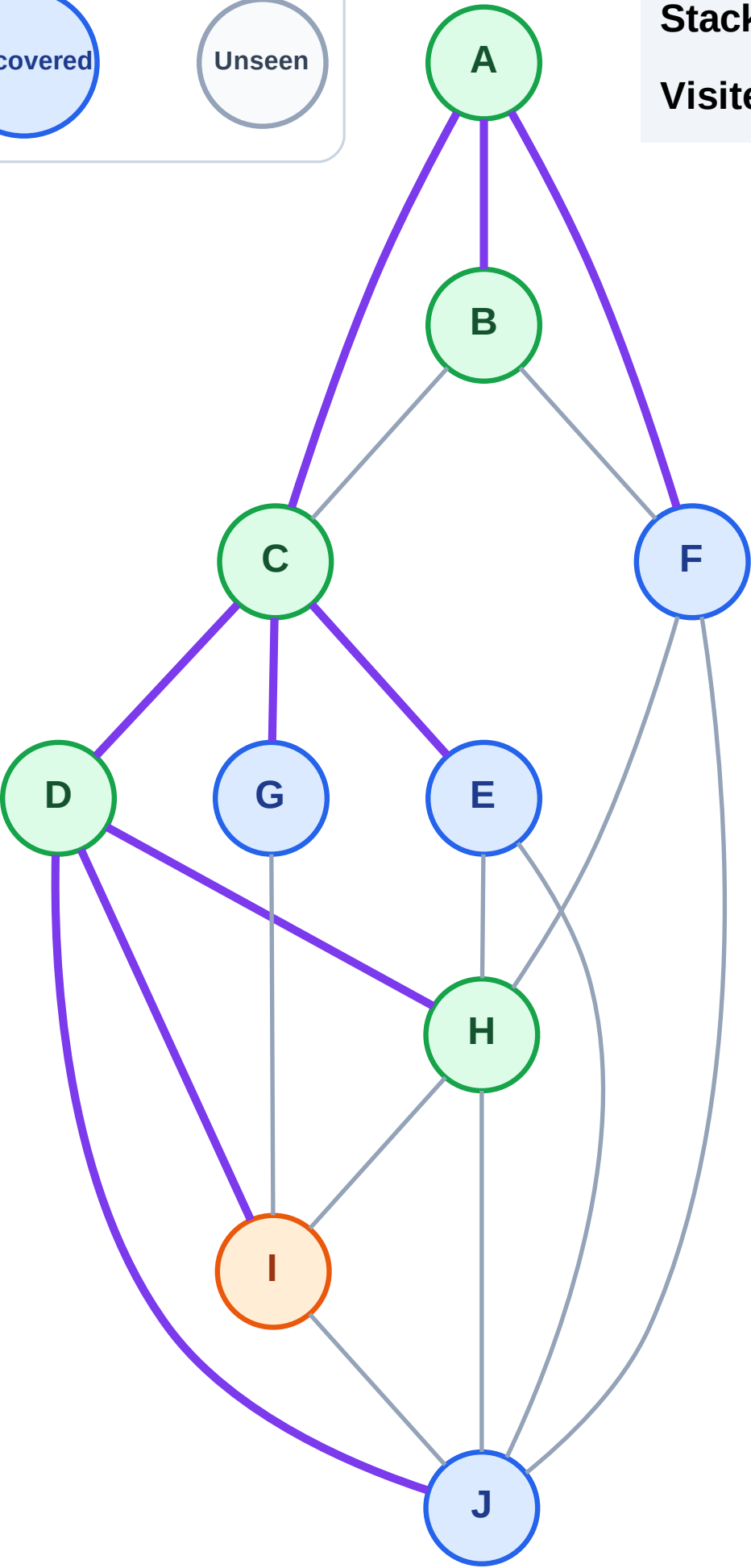
Processing

Discovered

Unseen

Stack (top at right): F | G | E | J

Visited: A, B, C, D, H



DFS Traversal

Step 13: No new neighbors from I

Legend

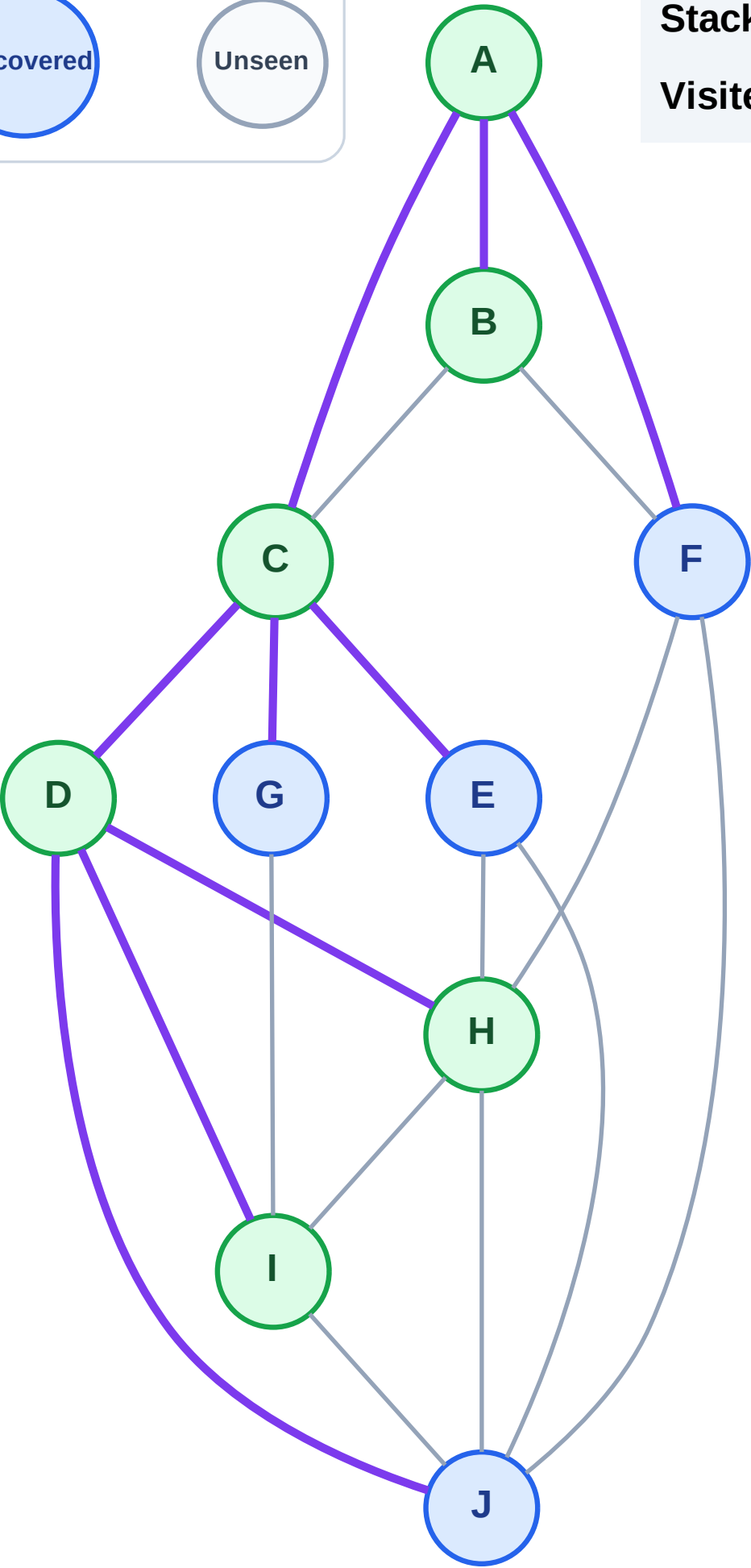
Complete

Processing

Discovered

Unseen

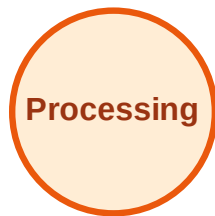
Stack (top at right): F | G | E | J
Visited: A, B, C, D, H, I



DFS Traversal

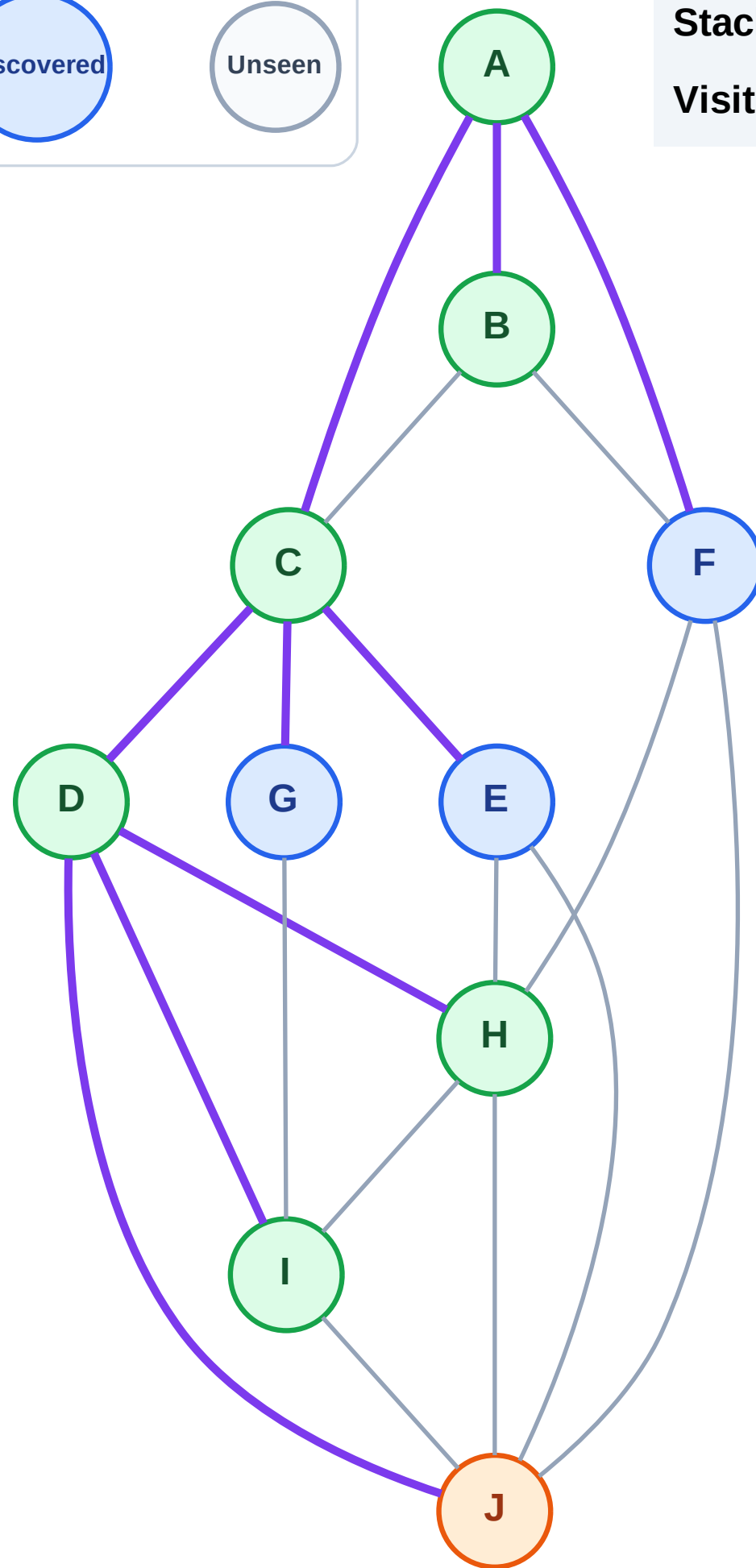
Step 14: Process node J

Legend



Stack (top at right): F | G | E

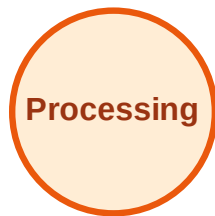
Visited: A, B, C, D, H, I



DFS Traversal

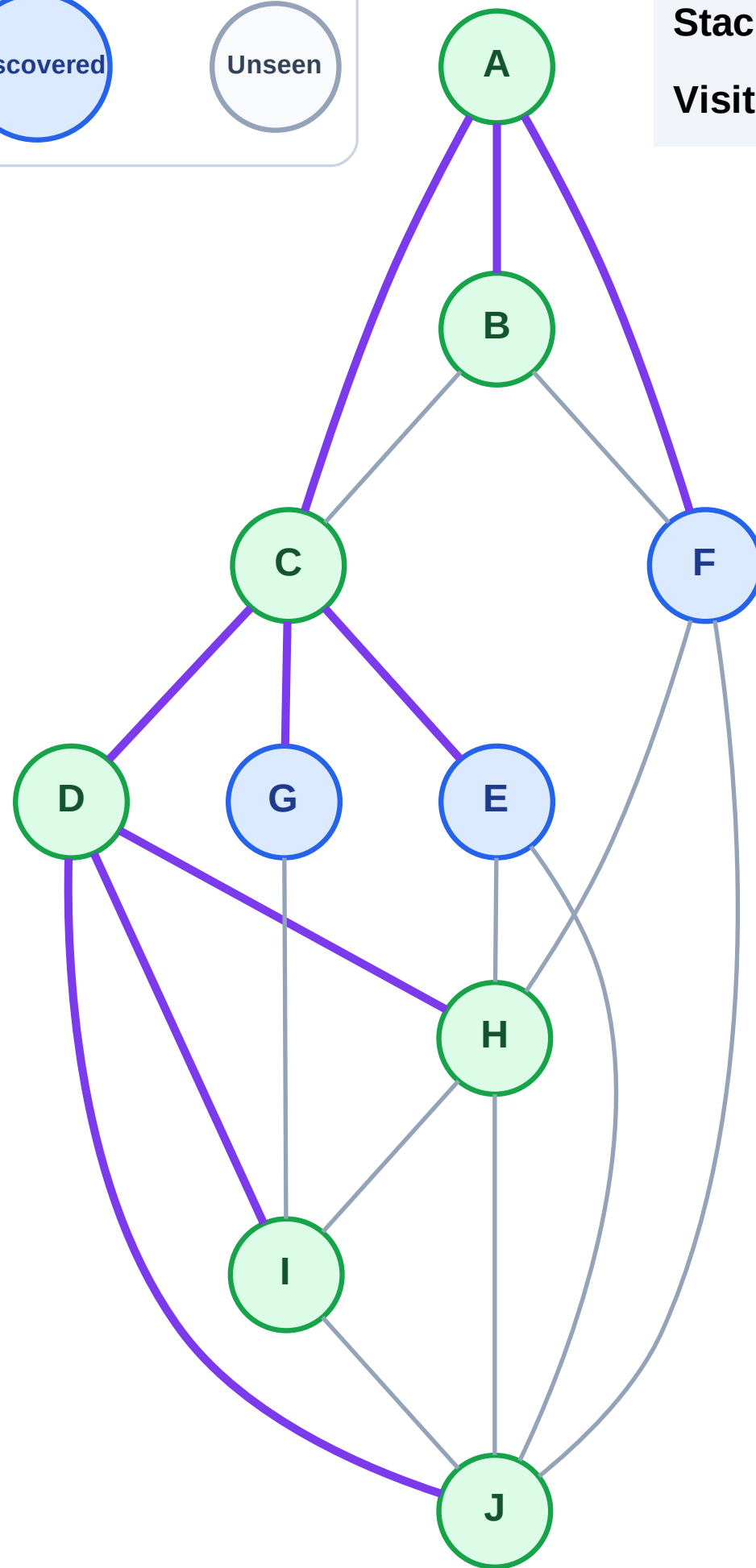
Step 15: No new neighbors from J

Legend



Stack (top at right): F | G | E

Visited: A, B, C, D, H, I, J



DFS Traversal

Step 16: Process node E

Legend

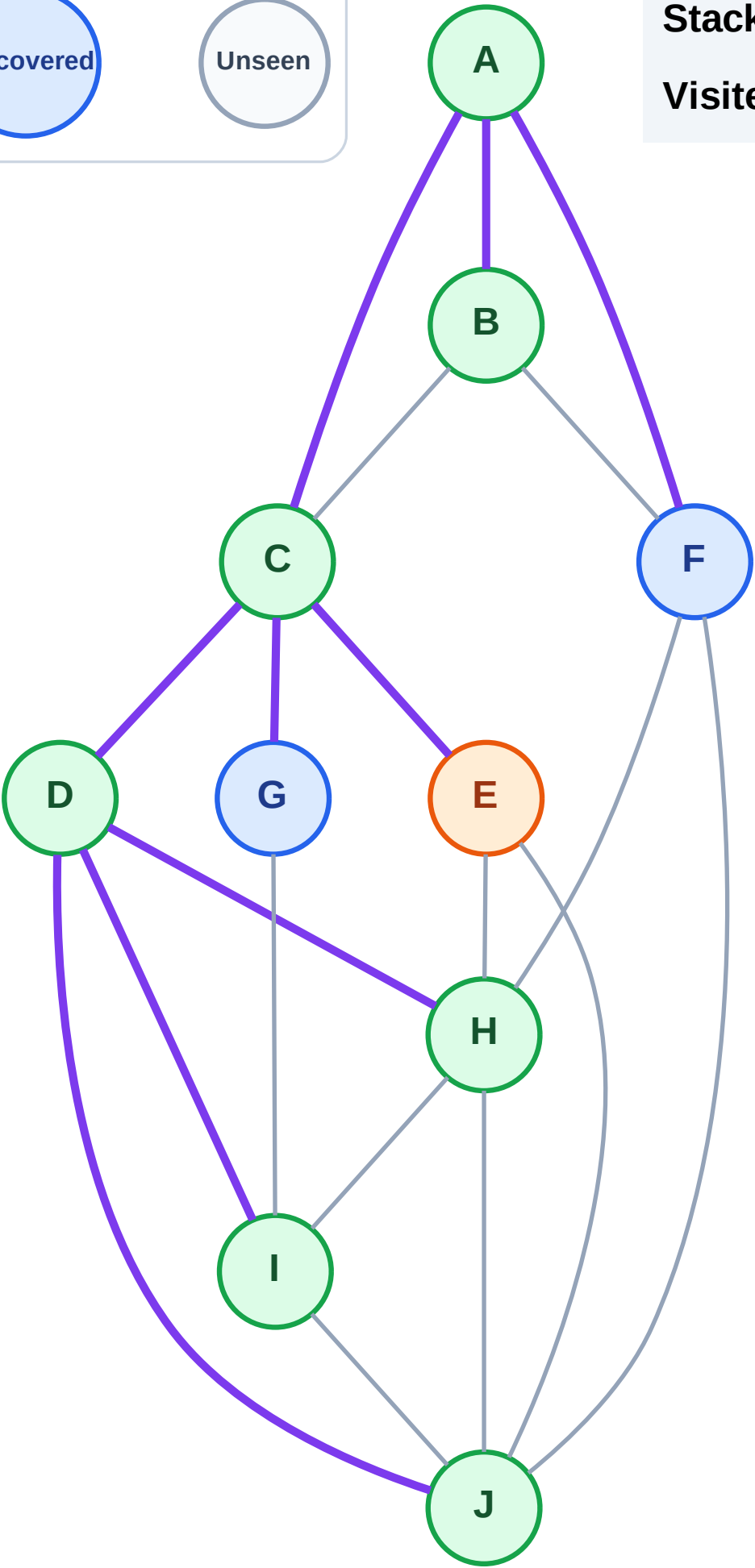
Complete

Processing

Discovered

Unseen

Stack (top at right): F | G
Visited: A, B, C, D, H, I, J



DFS Traversal

Step 17: No new neighbors from E

Legend

Complete

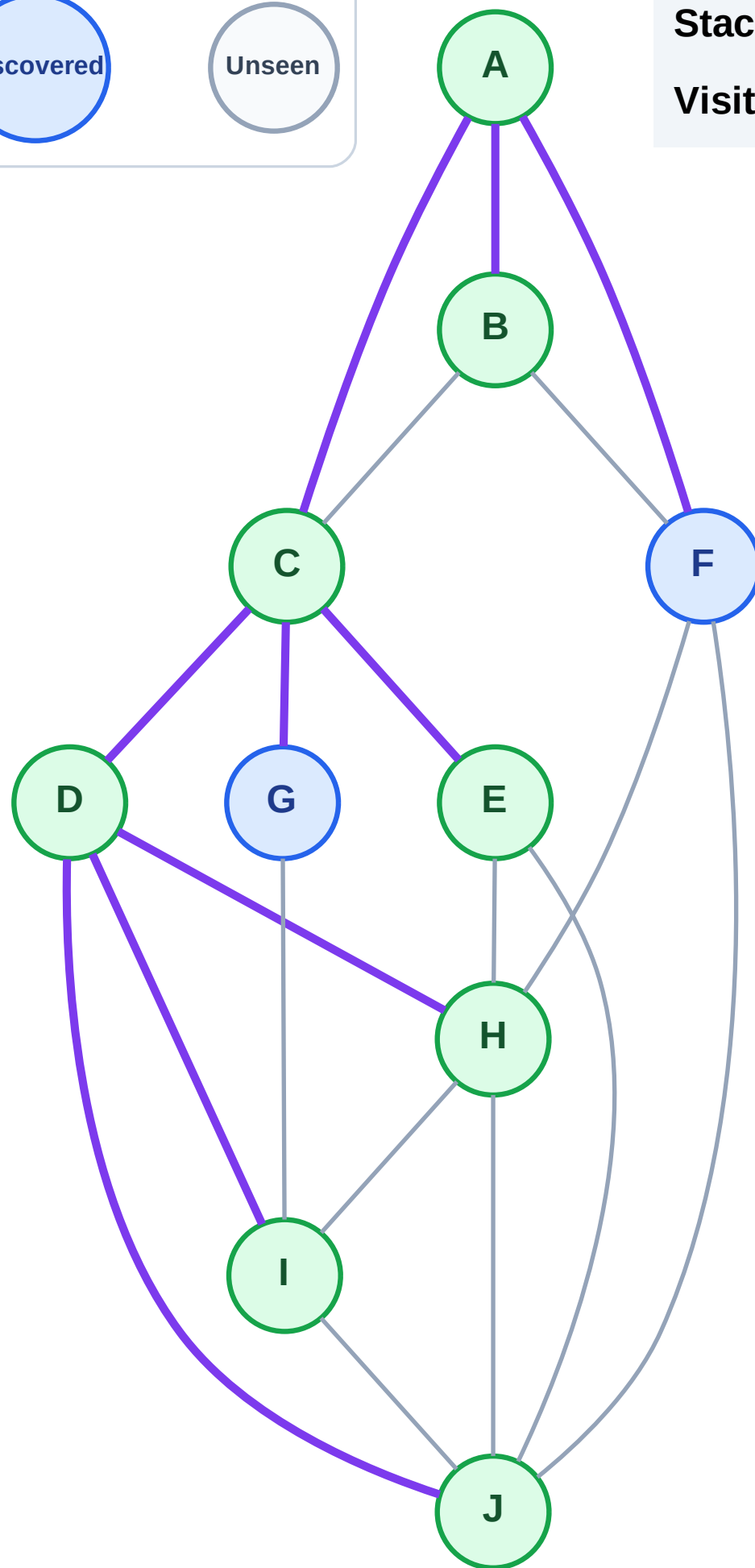
Processing

Discovered

Unseen

Stack (top at right): F | G

Visited: A, B, C, D, E, H, I, J



DFS Traversal

Step 18: Process node G

Legend

Complete

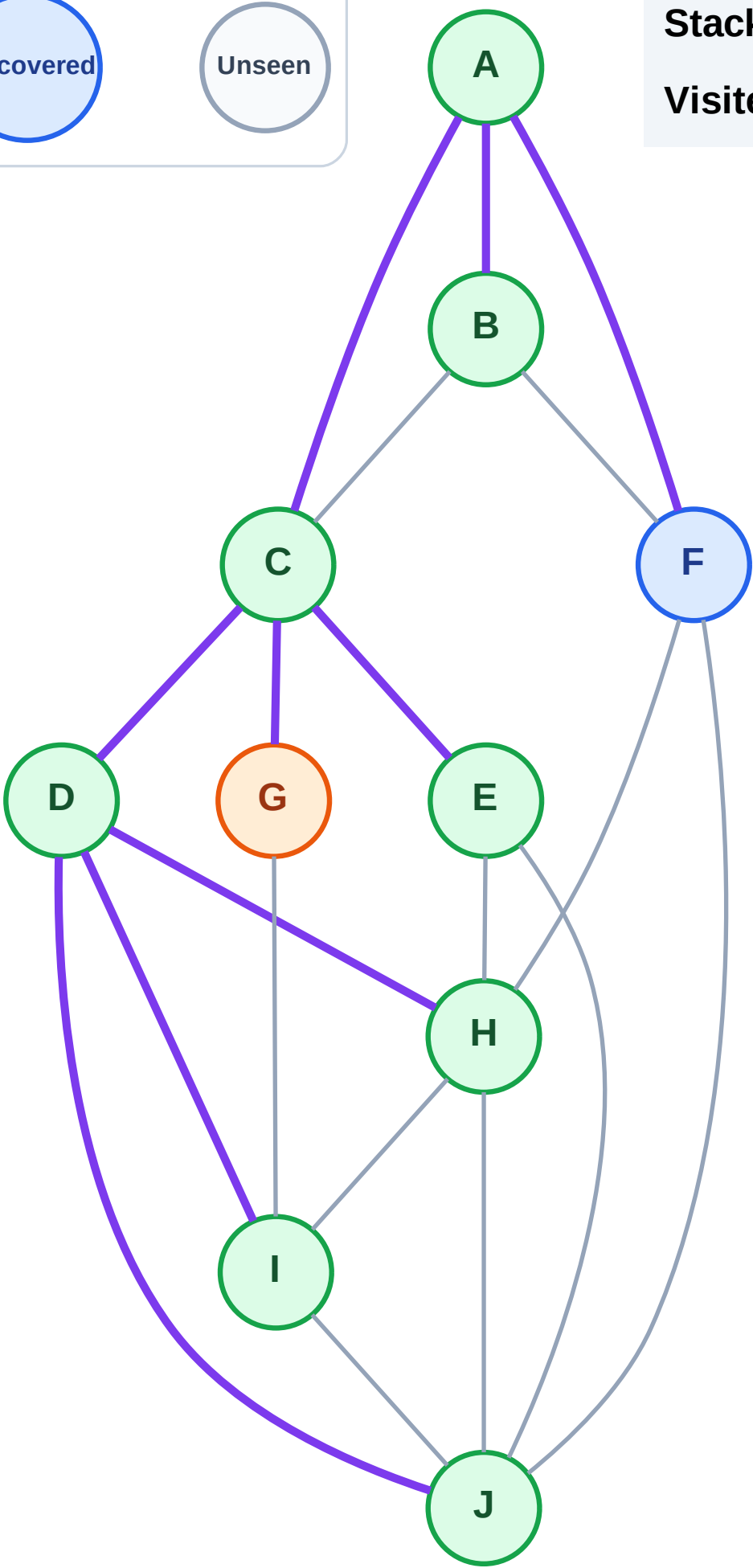
Processing

Discovered

Unseen

Stack (top at right): F

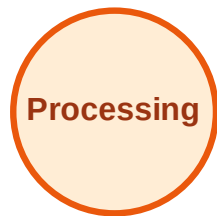
Visited: A, B, C, D, E, H, I, J



DFS Traversal

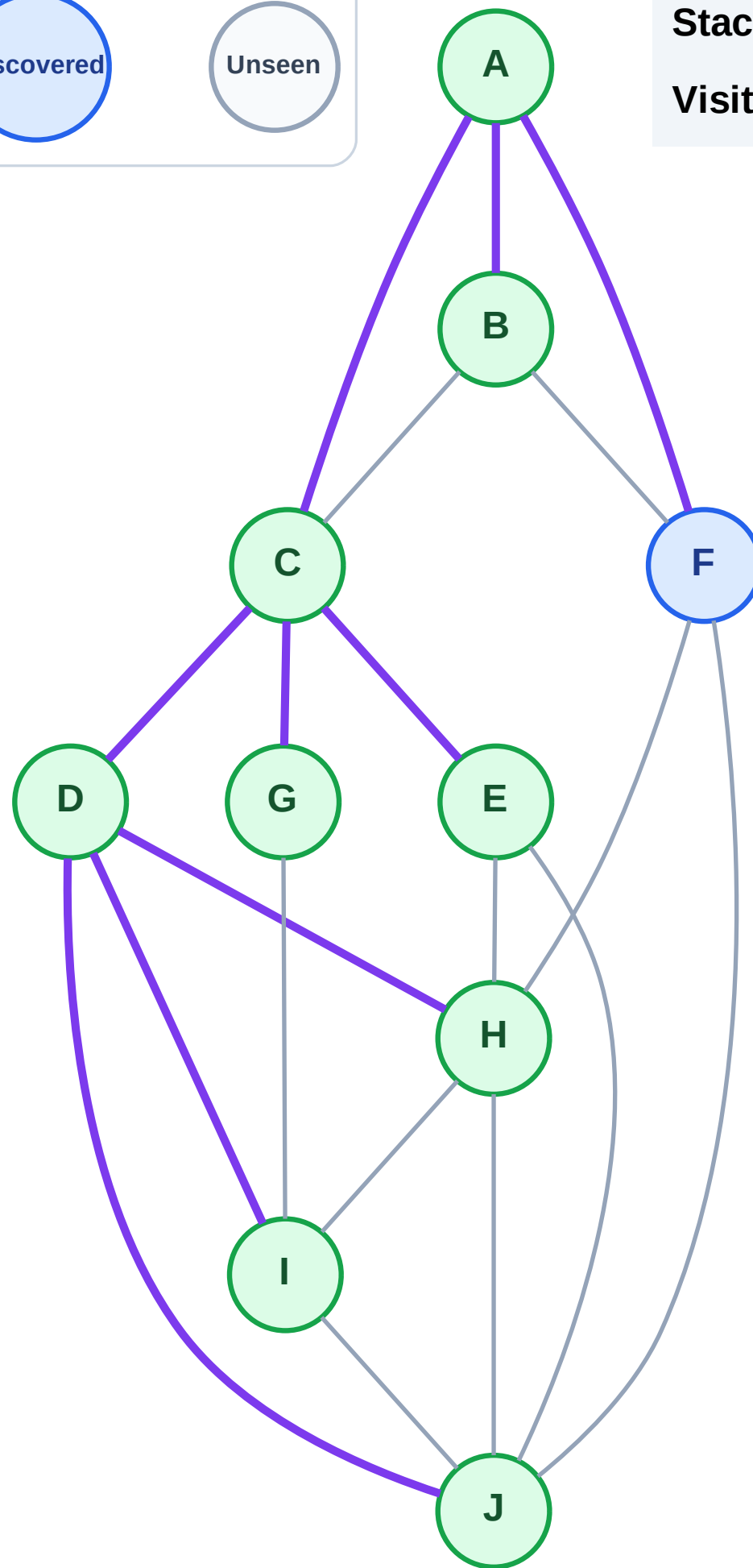
Step 19: No new neighbors from G

Legend



Stack (top at right): F

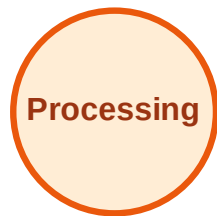
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

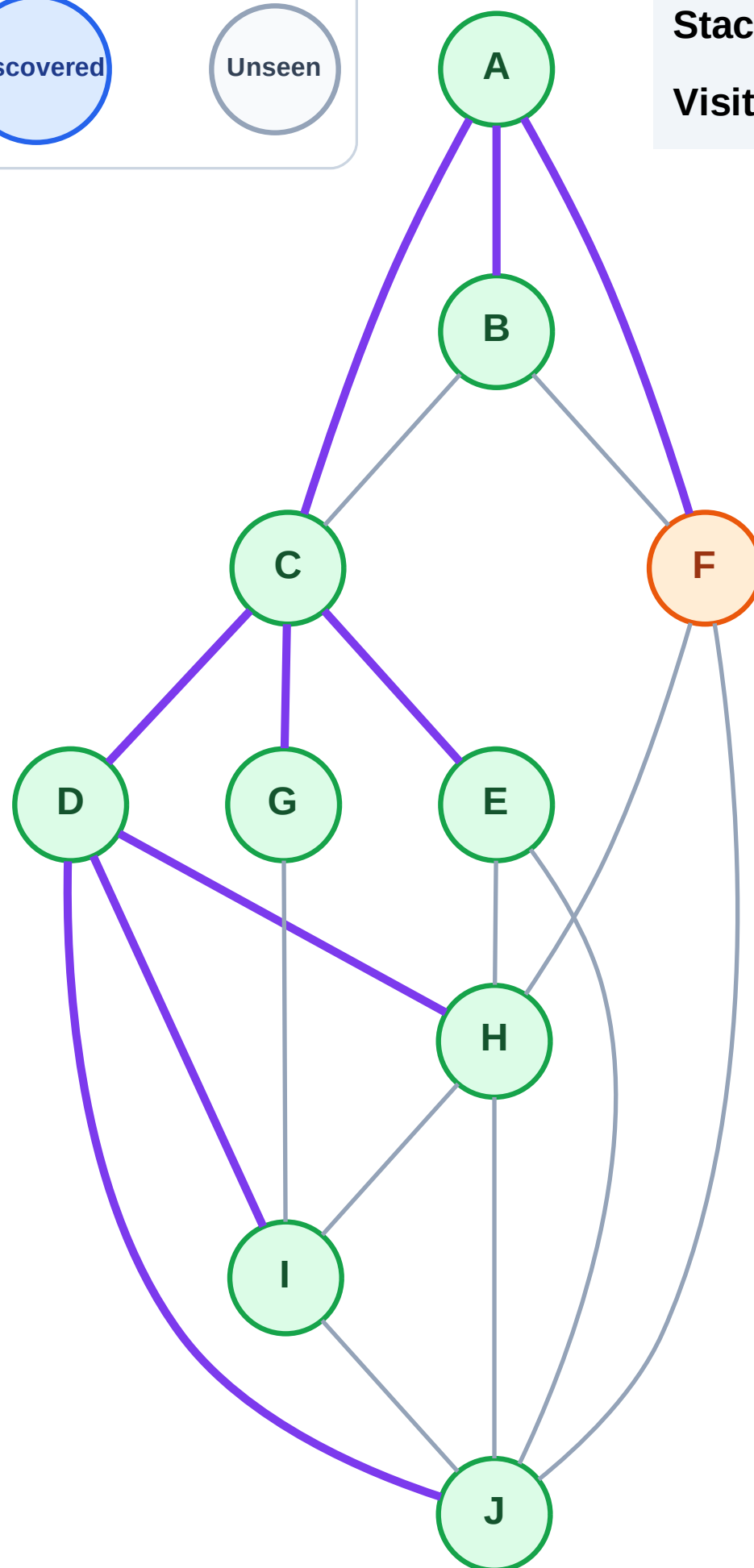
Step 20: Process node F

Legend



Stack (top at right): (empty)

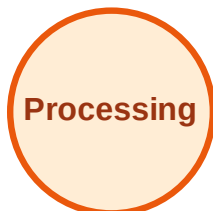
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

